



US012403537B2

(12) **United States Patent**
Sawa et al.

(10) **Patent No.:** **US 12,403,537 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **HOLE SAW**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/606,157**

(22) Filed: **Mar. 15, 2024**

(65) **Prior Publication Data**

US 2024/0217008 A1 Jul. 4, 2024

Related U.S. Application Data

(60) Continuation of application No. 18/065,305, filed on Dec. 13, 2022, which is a division of application No. (Continued)

(51) **Int. Cl.**
B23B 51/04 (2006.01)

(52) **U.S. Cl.**
CPC **B23B 51/0467** (2022.01); **B23B 51/0466** (2013.01); **B23B 51/0426** (2013.01); **B23B 2260/072** (2013.01); **Y10T 408/895** (2015.01)

(58) **Field of Classification Search**
CPC B23B 51/0426; B23B 51/0466; B23B 51/0467; Y10T 408/895

See application file for complete search history.

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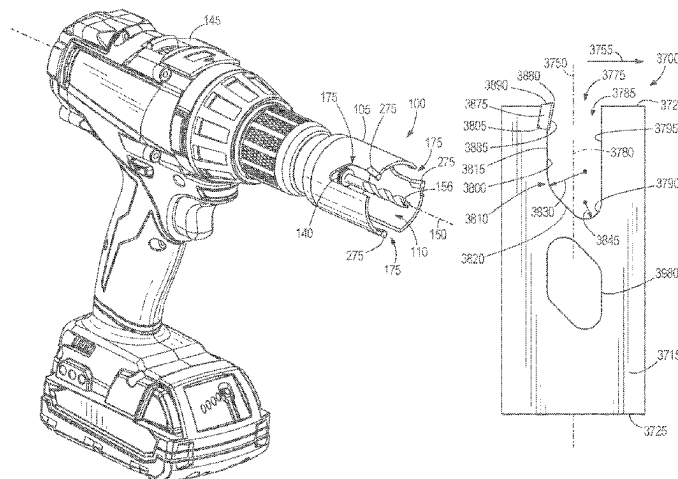
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(57) **ABSTRACT**

A hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body with a sidewall having a first end and a second end opposite the first end, a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall, and a support aperture formed in the sidewall. The gullet extends through the first end of the body. The support aperture is entirely surrounded by the sidewall. A majority of the support aperture circumferentially overlaps with the gullet. At least a portion of the support aperture is positioned closer than the bottom surface of the sidewall to the second end. The hole saw includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet and an end cap coupled to the second end of the sidewall.

20 Claims, 12 Drawing Sheets



Related U.S. Application Data

16/910,166, filed on Jun. 24, 2020, now Pat. No. 11,559,840, which is a division of application No. 15/863,253, filed on Jan. 5, 2018, now Pat. No. 10,730,119.

- (60) Provisional application No. 62/443,317, filed on Jan. 6, 2017.

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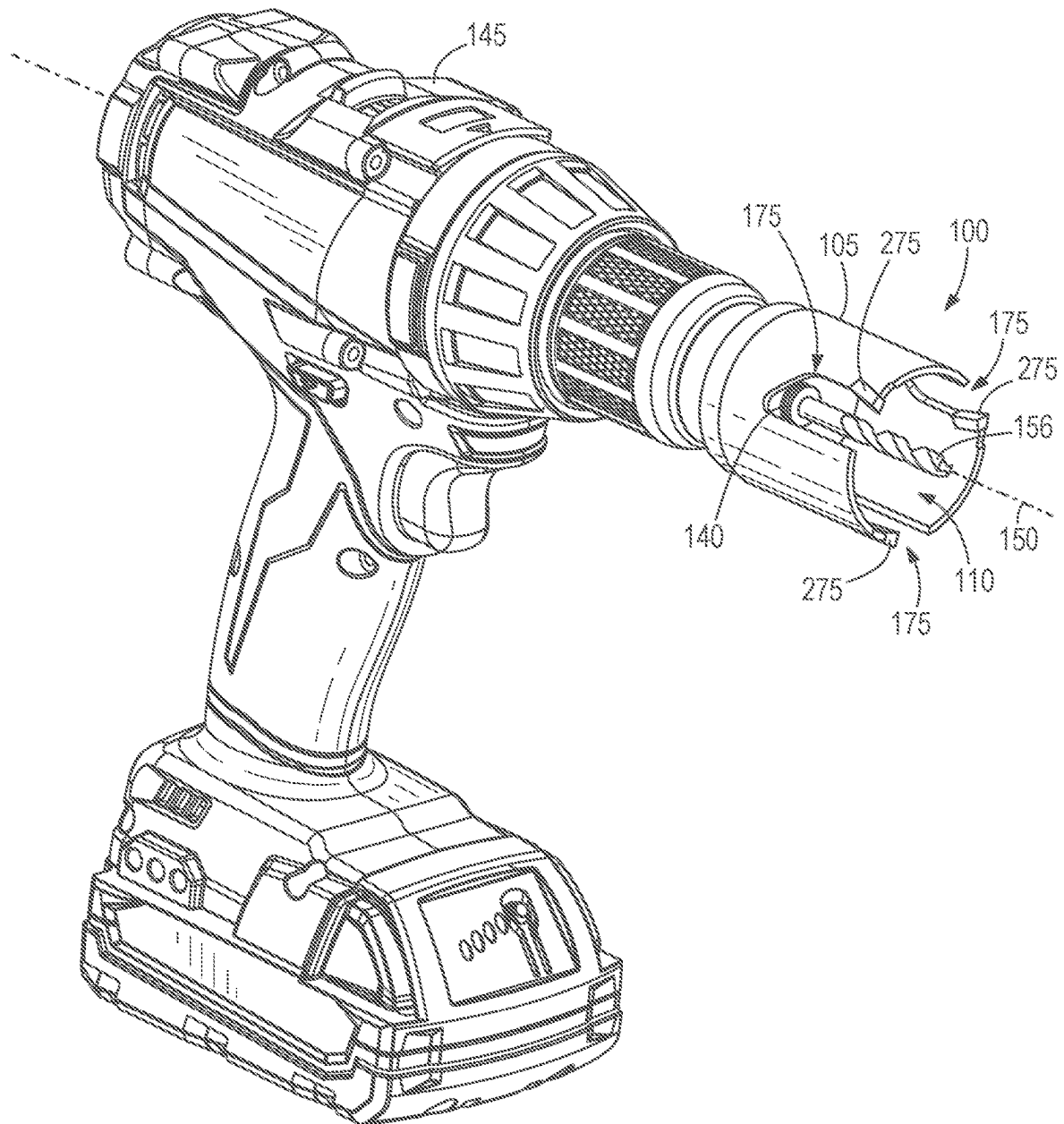
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**FIG. 1**

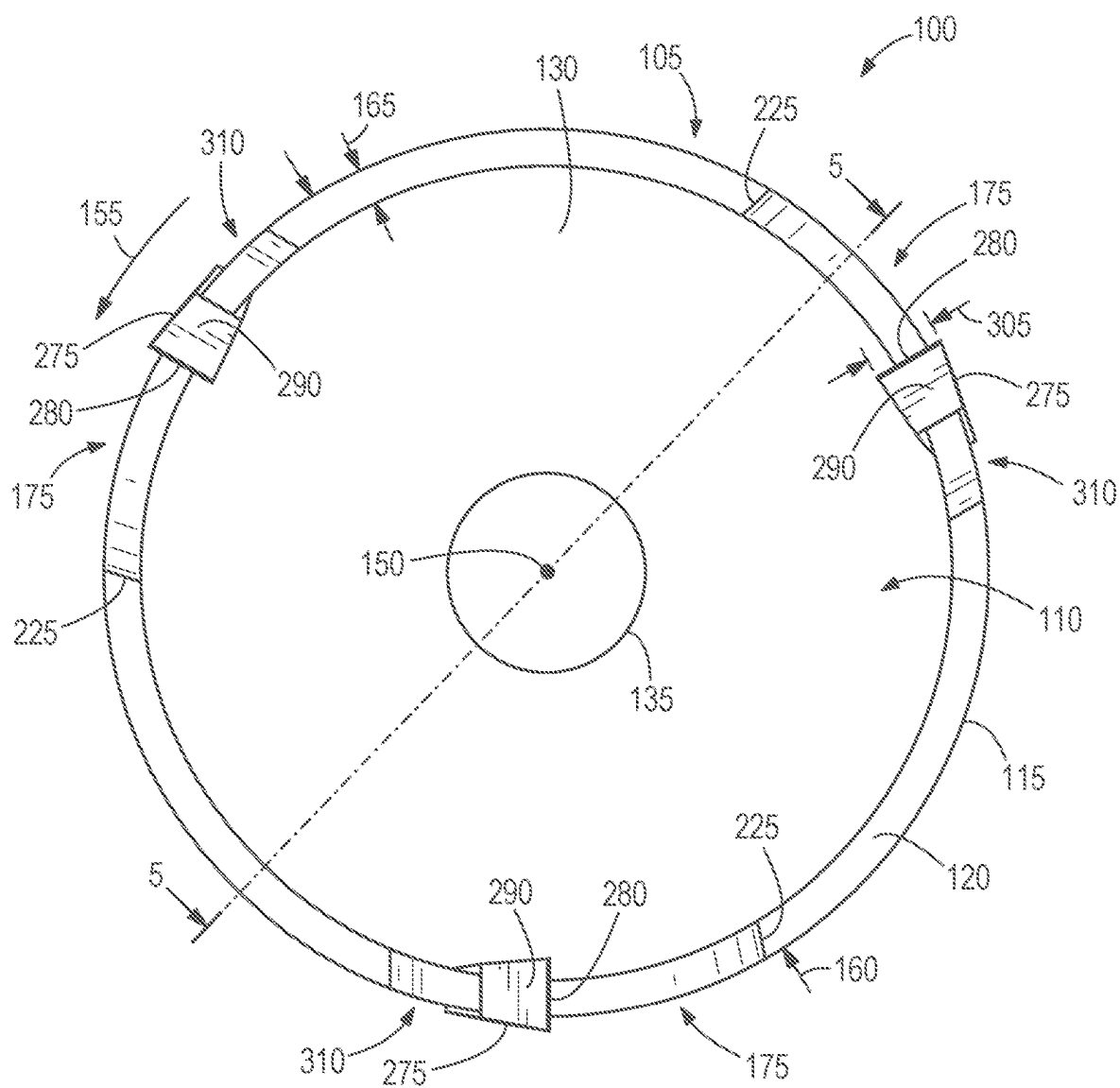


FIG. 2

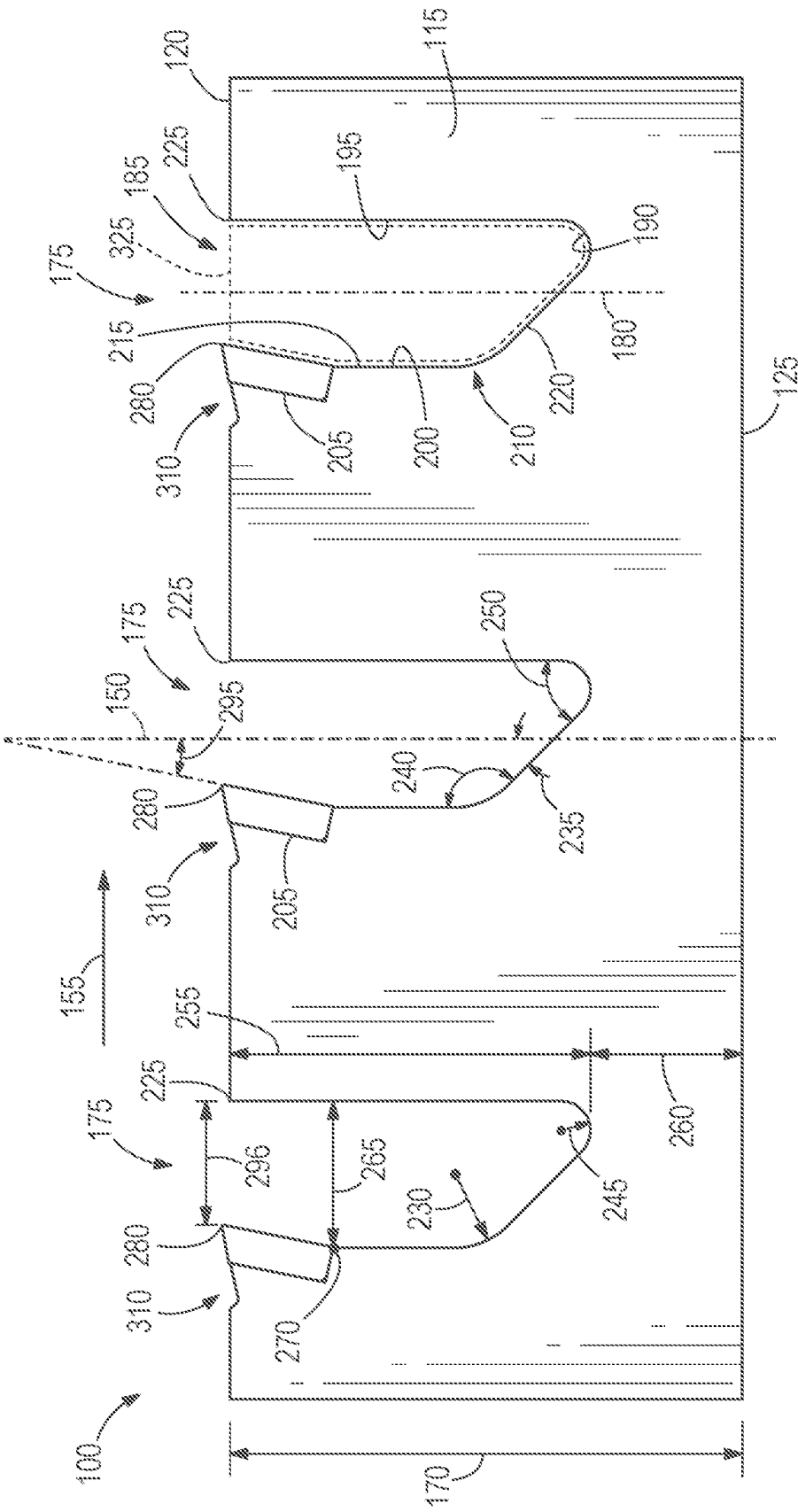


FIG. 3

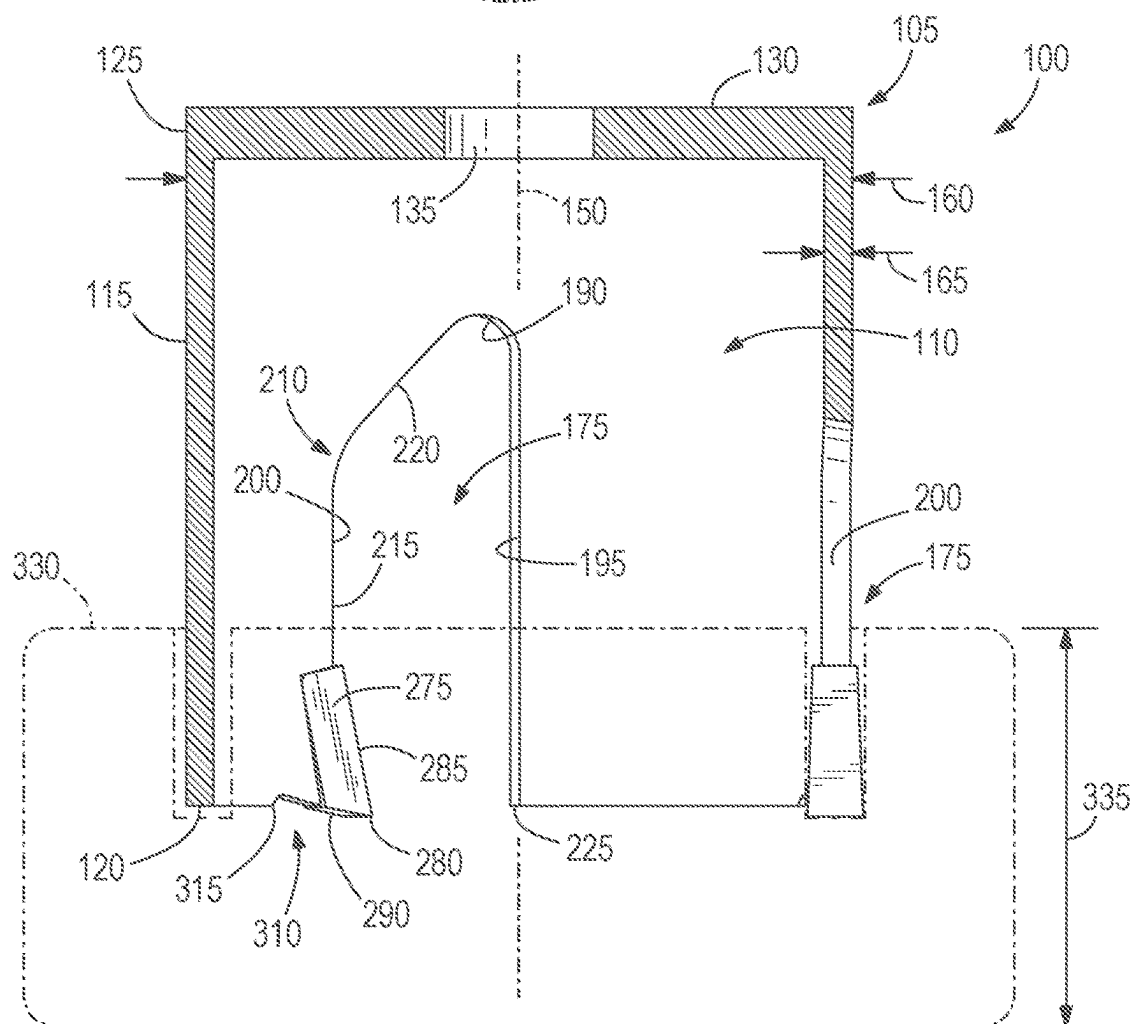
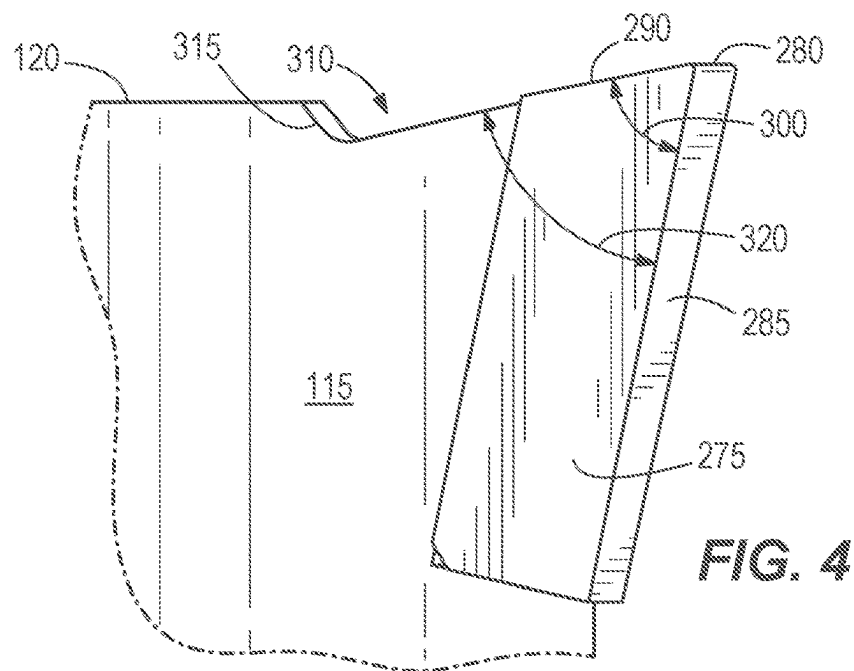


FIG. 5

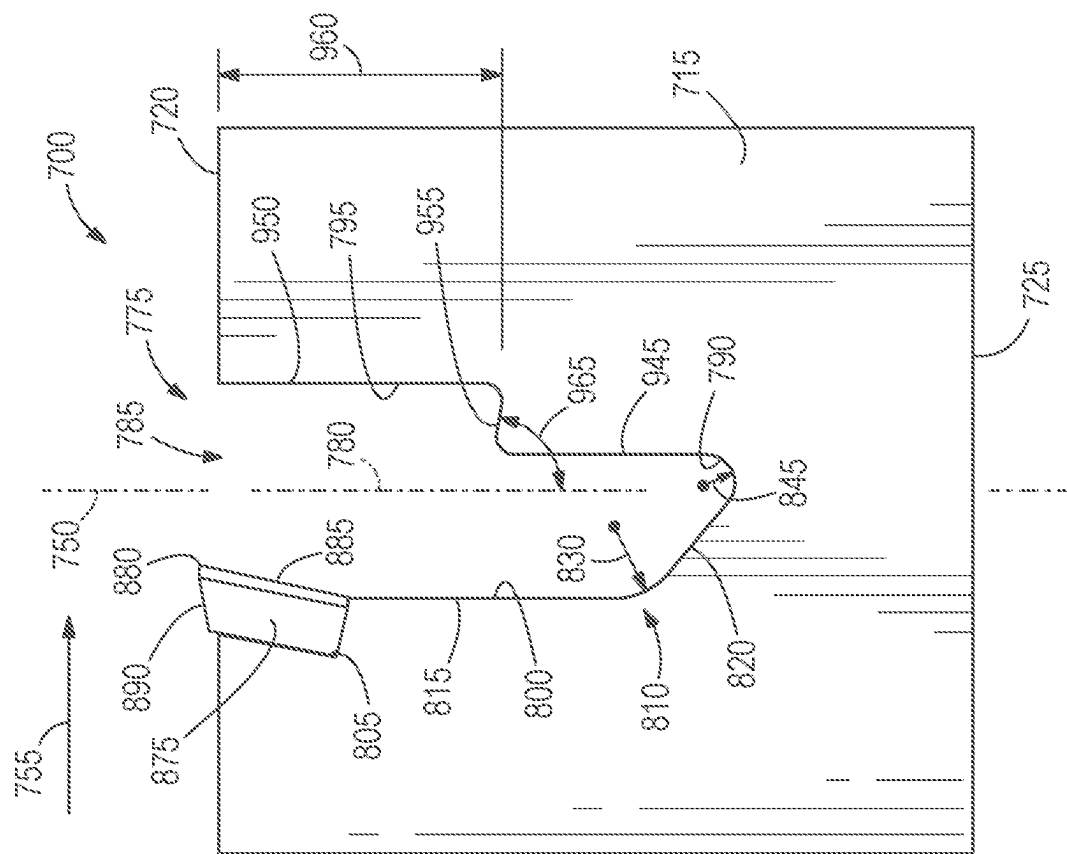


FIG. 6

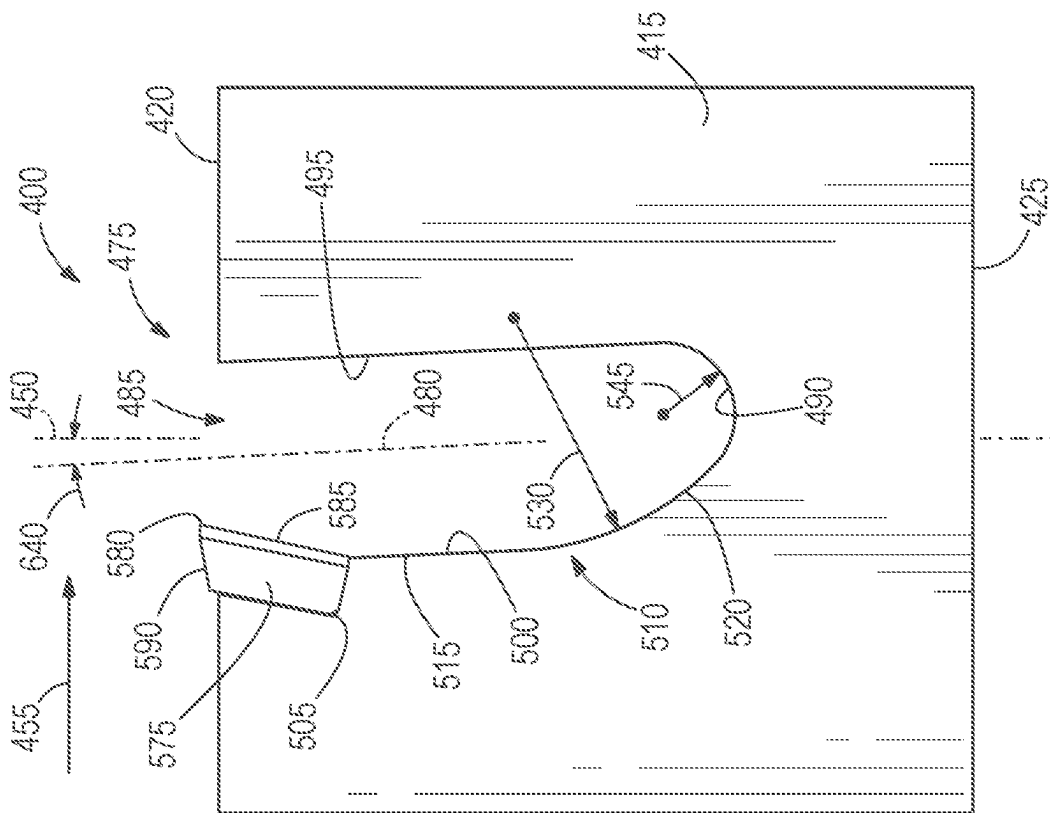
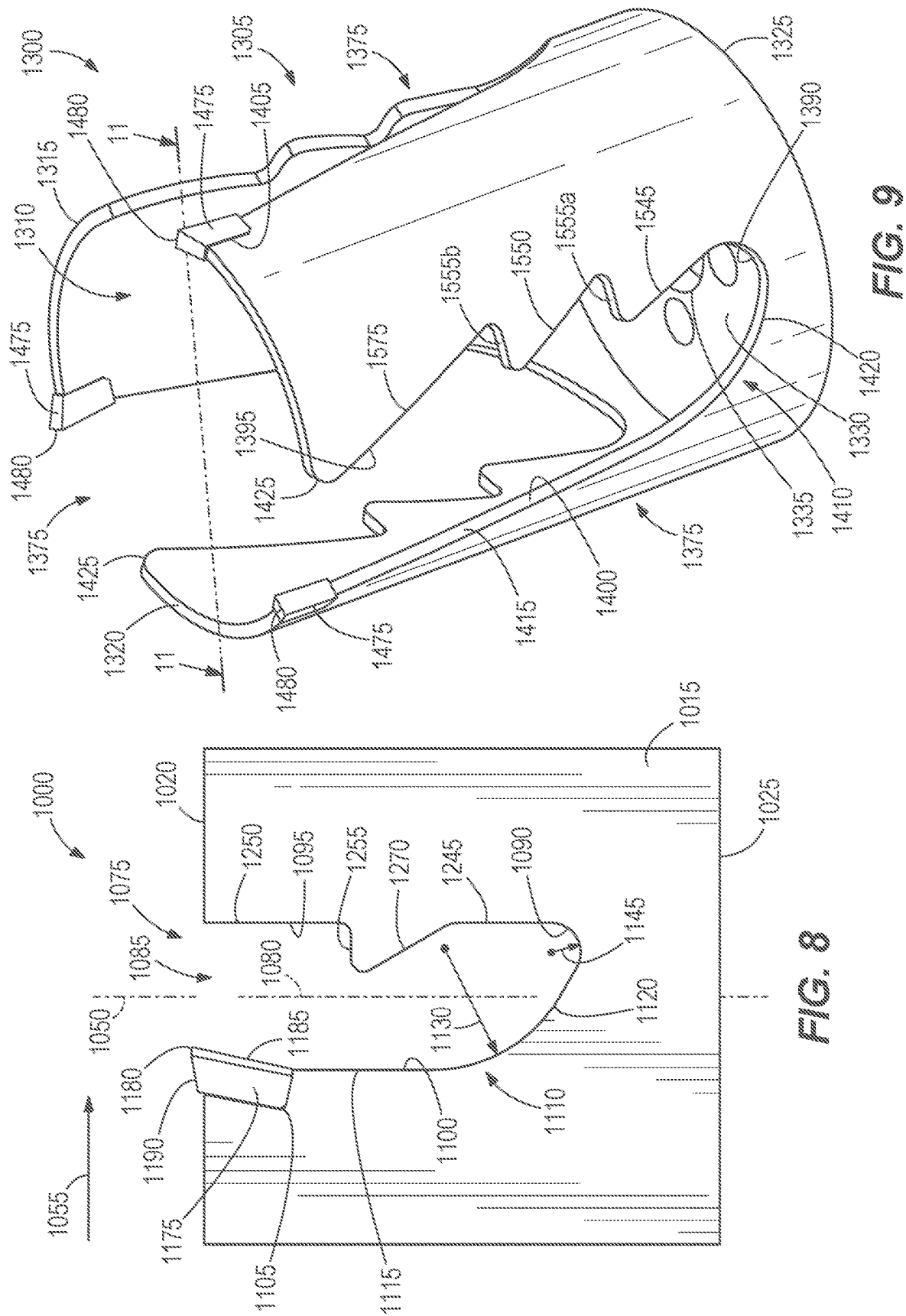


FIG. 7



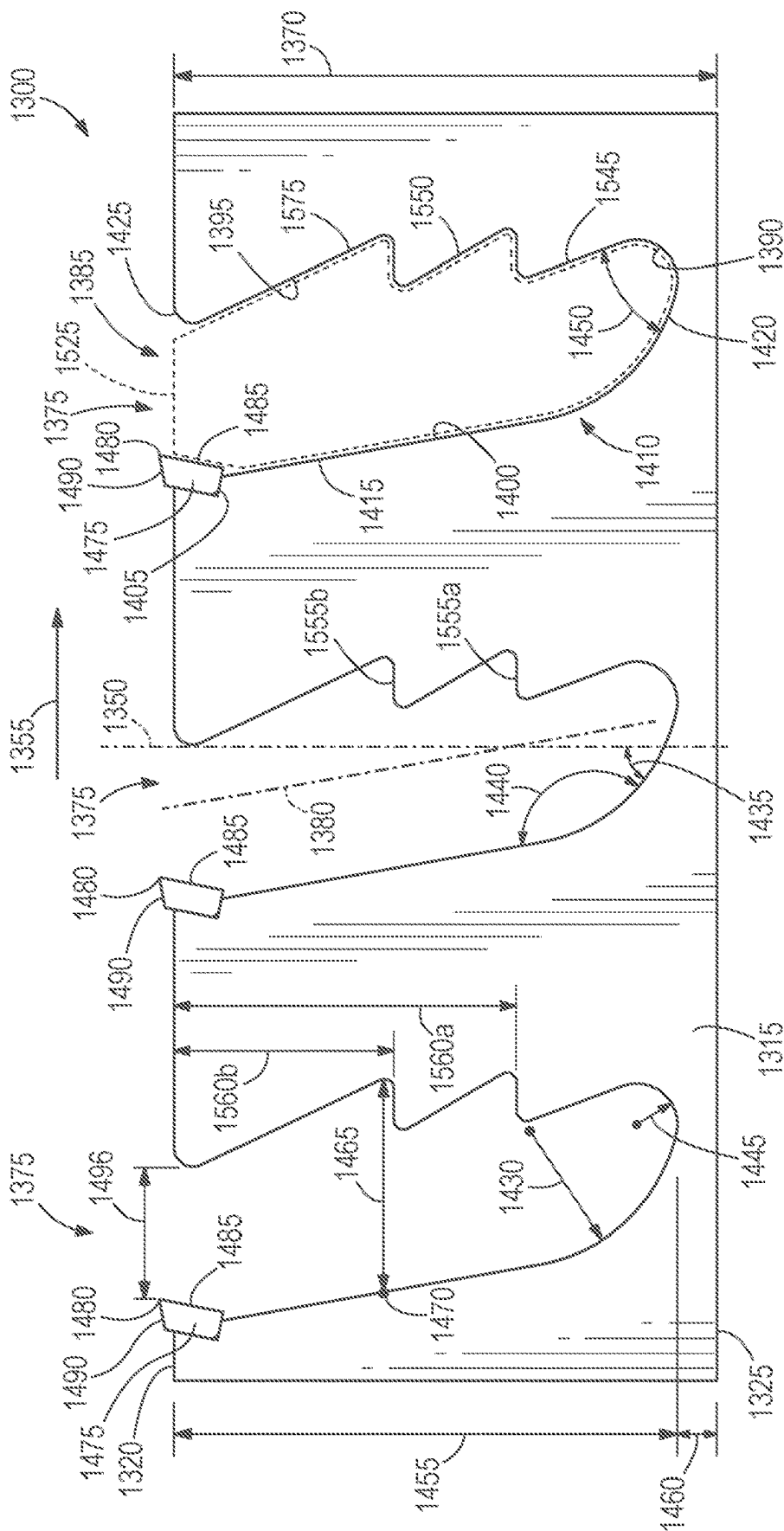


FIG. 10

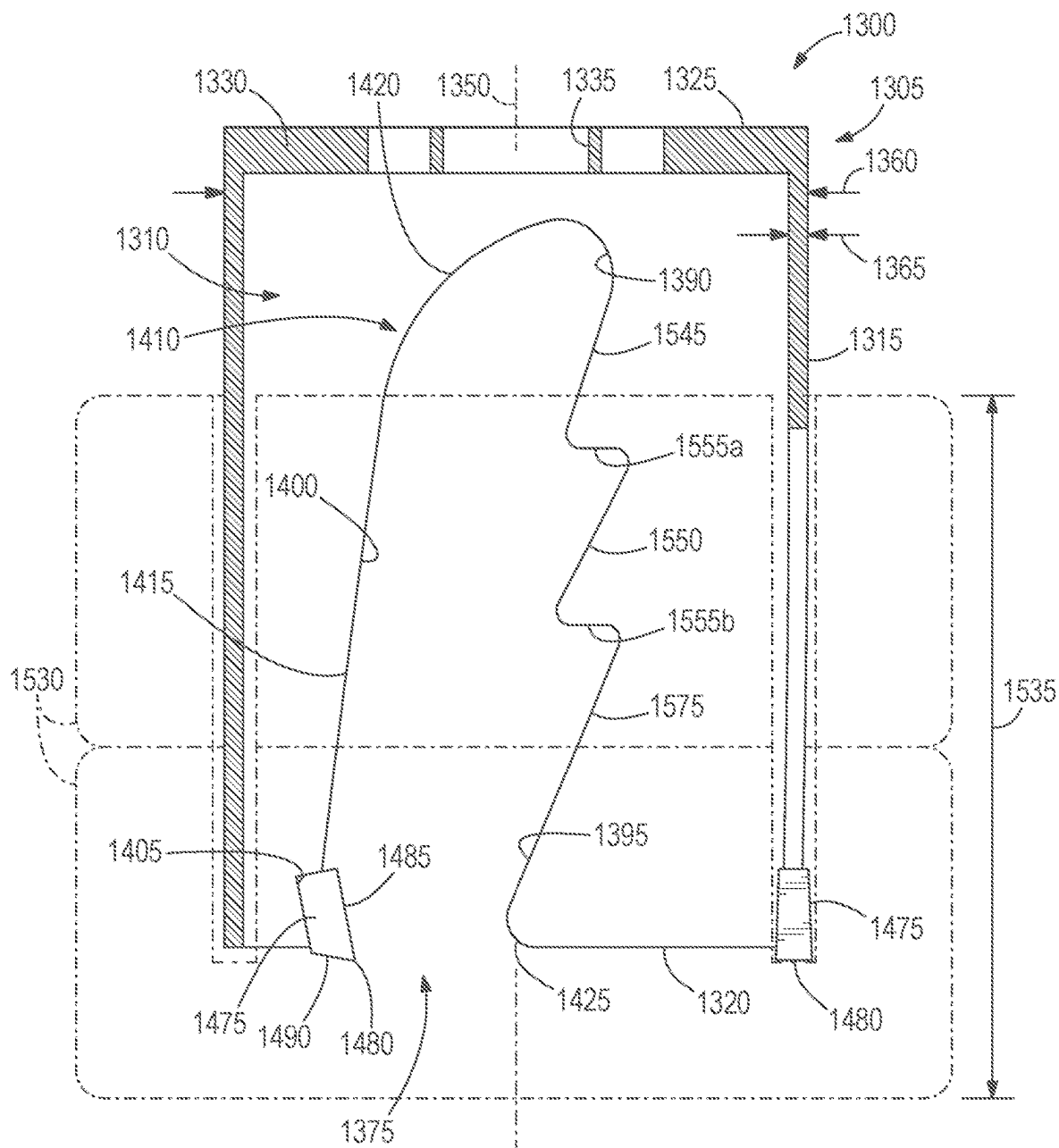


FIG. 11

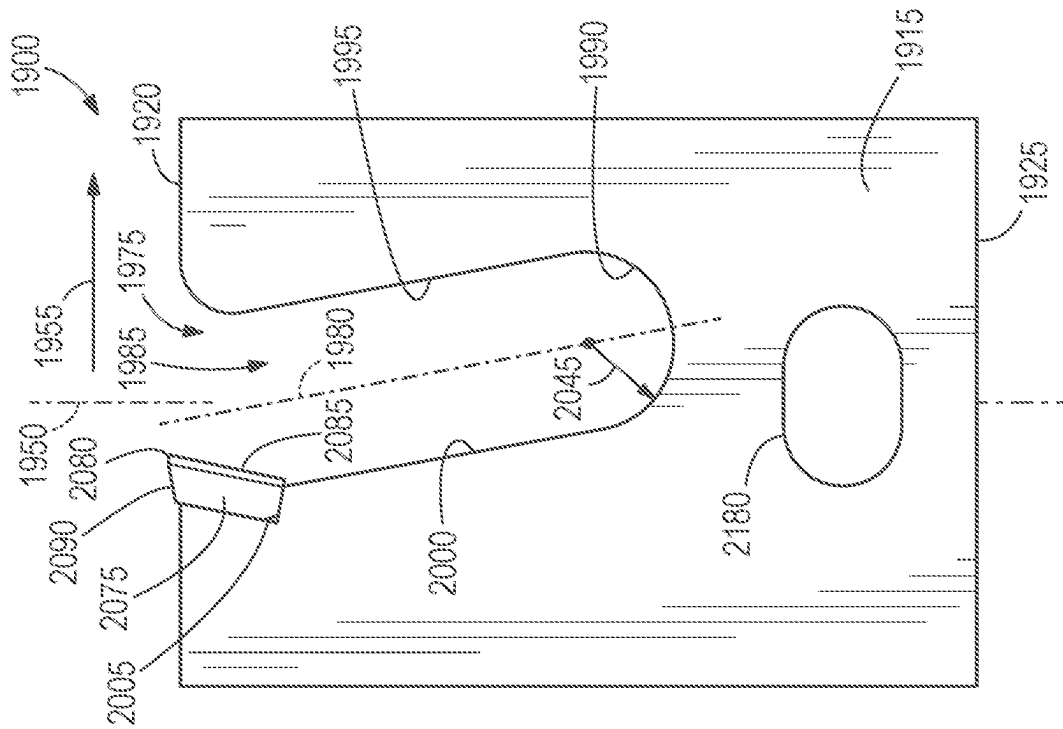


Fig. 13

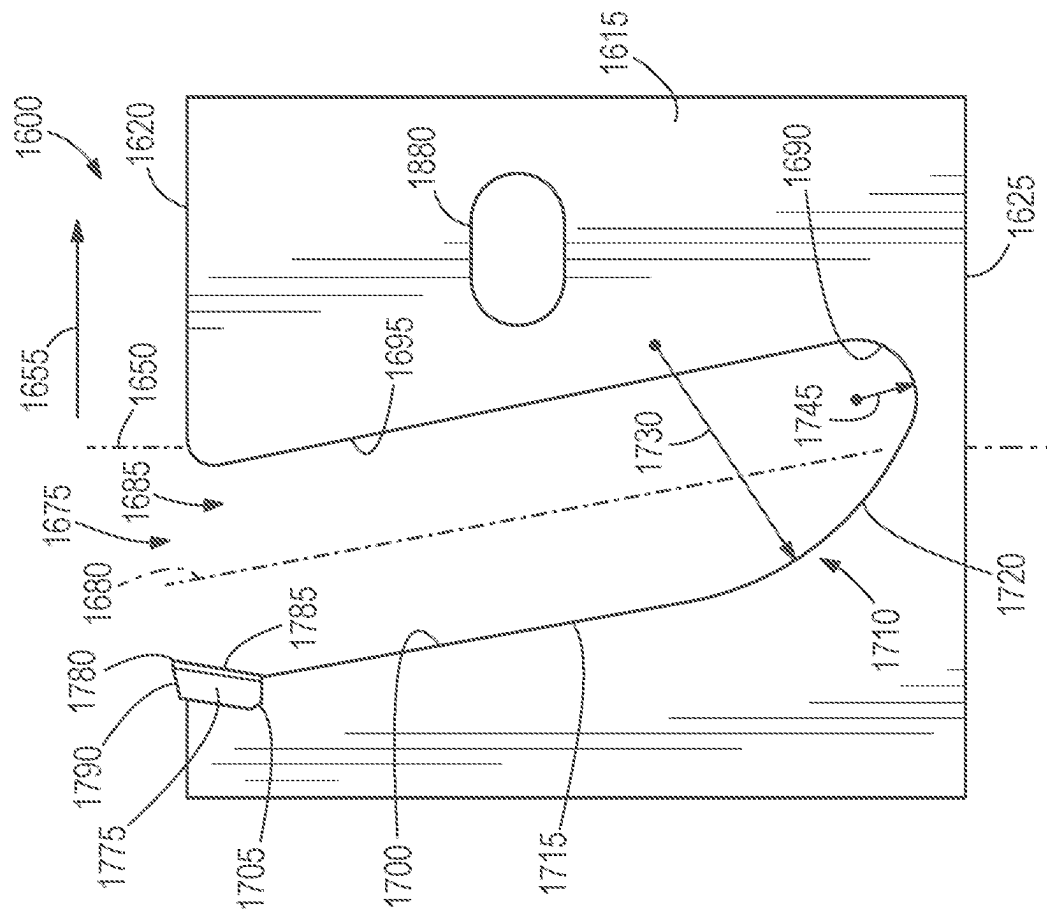


FIG. 12

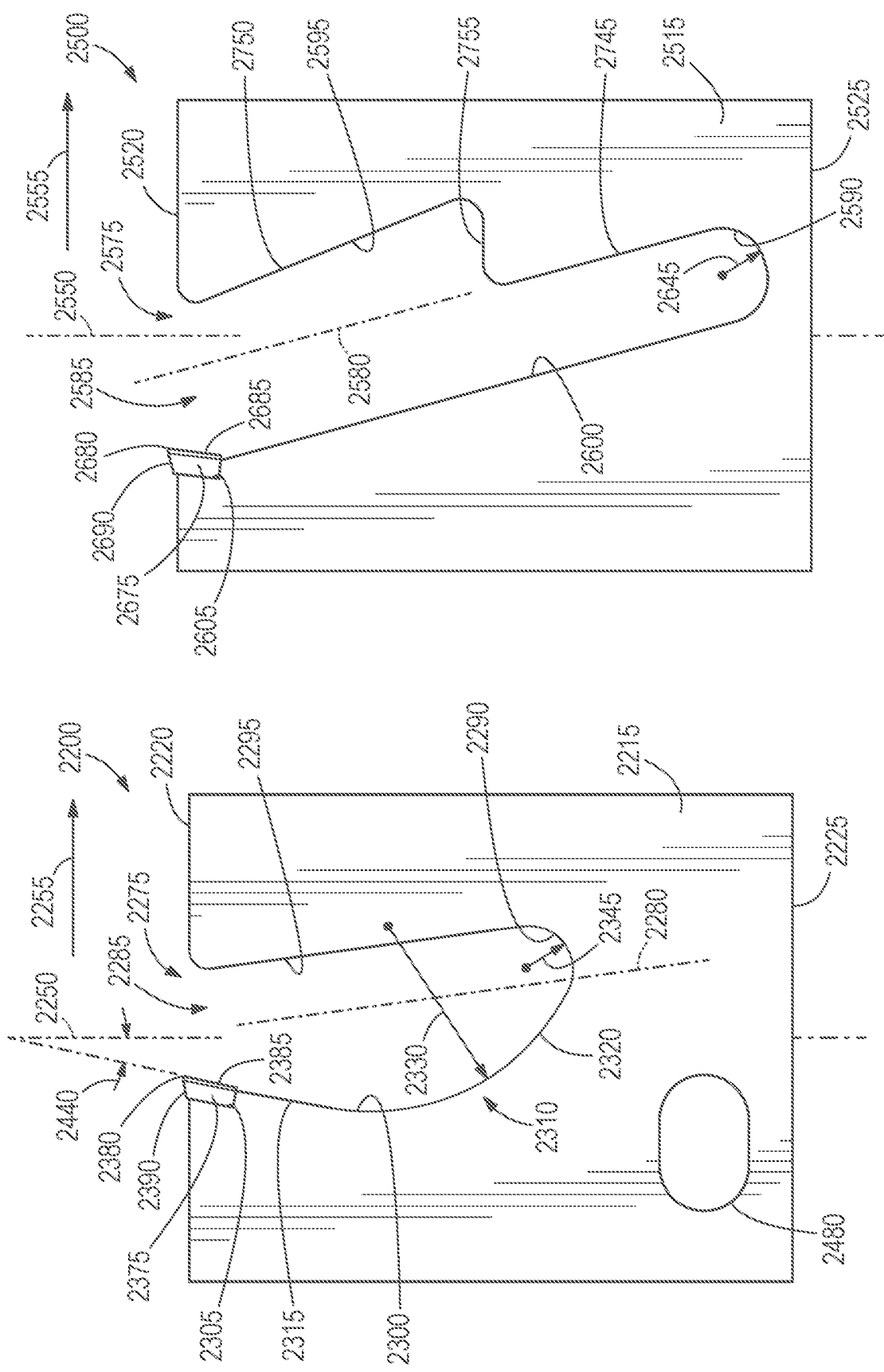
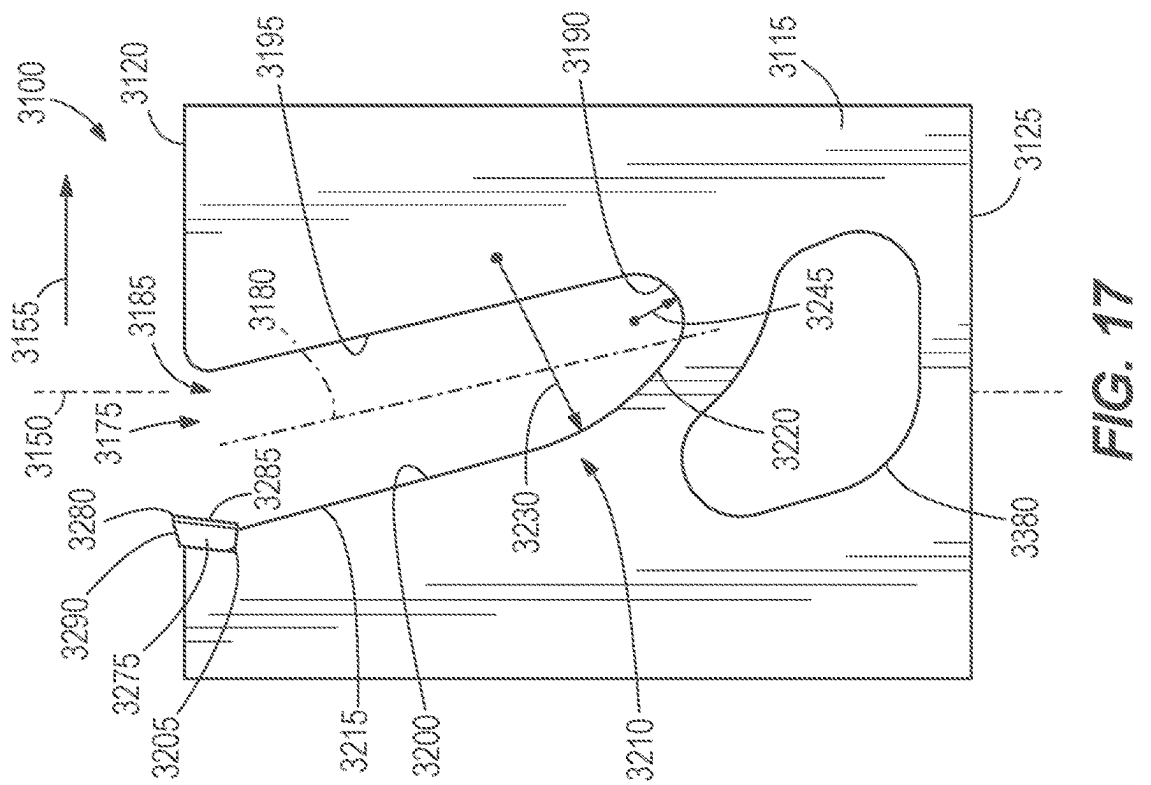
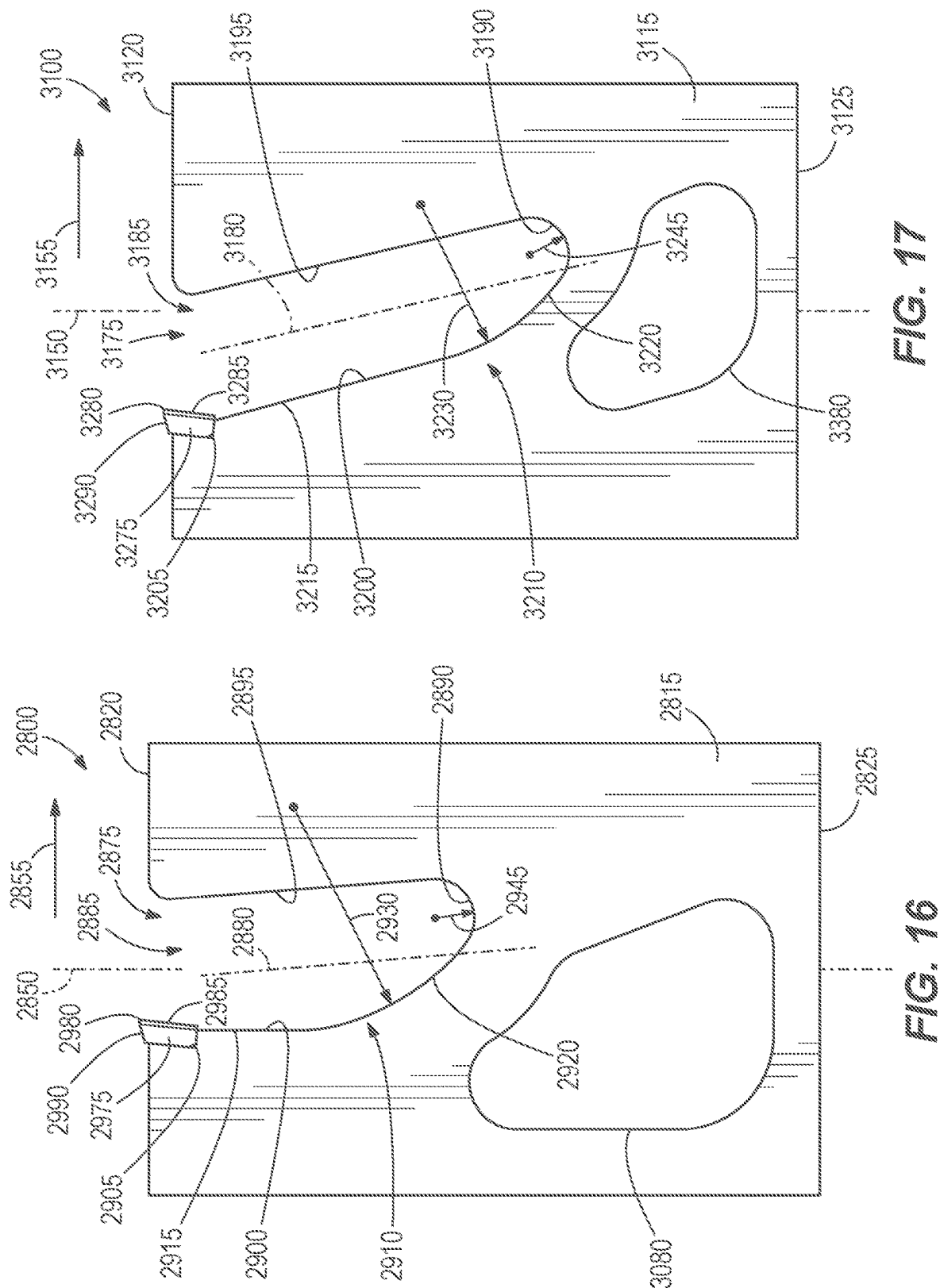
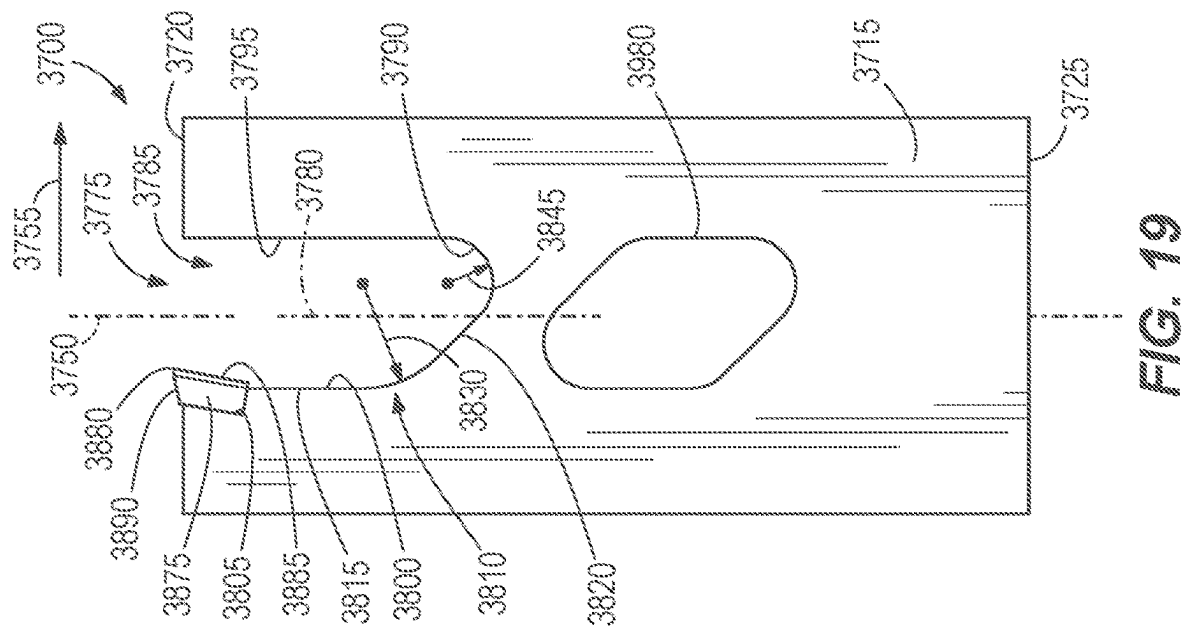


FIG. 15

FIG. 14





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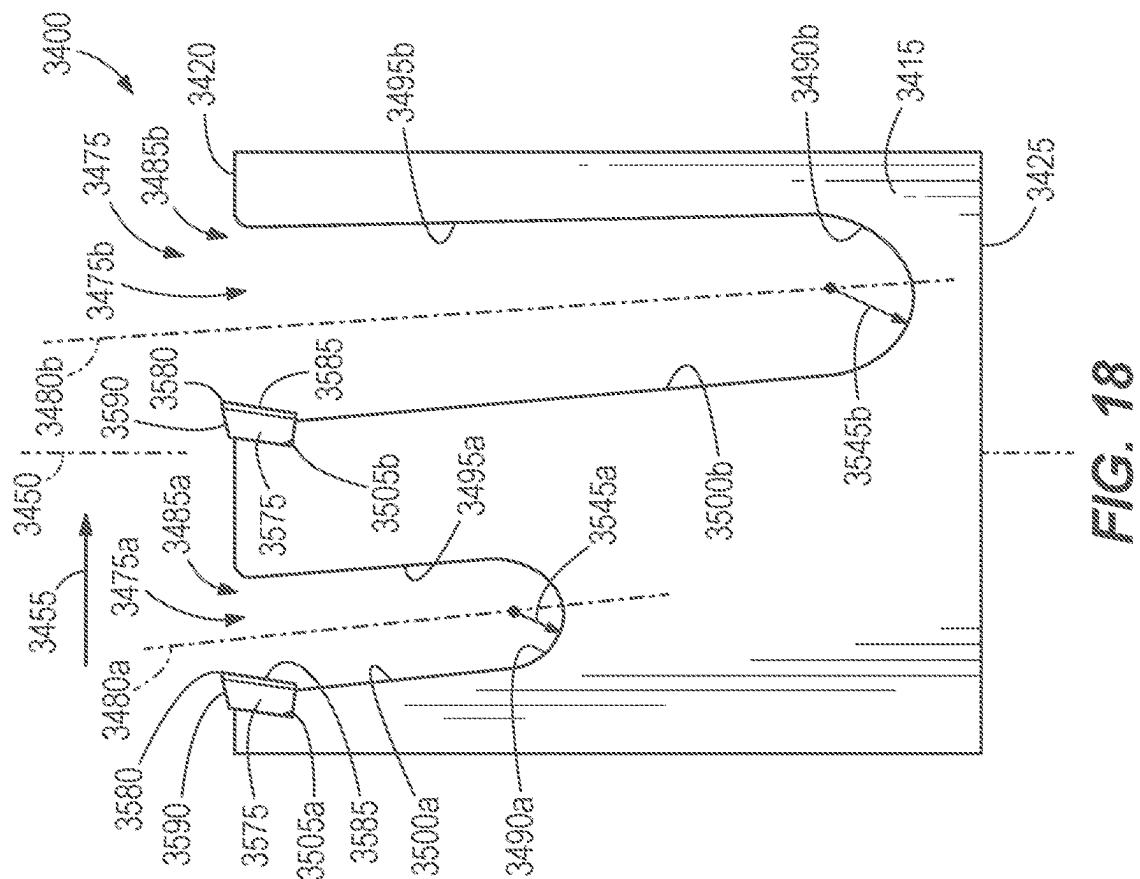


Fig. 18

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HOLE SAW**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 18/065,305, filed Dec. 13, 2022, now U.S. Pat. No. 12,233,467, which is a divisional of U.S. patent application Ser. No. 16/910,166, filed Jun. 24, 2020, now U.S. Pat. No. 11,559,840, which is a divisional of U.S. patent application Ser. No. 15/863,253, filed Jan. 5, 2018, now U.S. Pat. No. 10,730,119, which claims priority to U.S. Provisional Patent Application No. 62/443,317, filed Jan. 6, 2017, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to hole saws, and more particularly to gullets formed in the hole saws.

SUMMARY

In one aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The gullet defines a central longitudinal axis extending between the first and second ends of the sidewall that is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The gullet has a dimension measured parallel to the rotational axis between the first end of the sidewall and the bottom surface of the gullet that is greater than 1.5 inches. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet.

In another aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The gullet extends through the first end of the sidewall and defines a central longitudinal axis extending between the first and second ends of the sidewall. The central longitudinal axis is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet. A dimension measured between the leading surface and a tip of the cutting tooth in a direction perpendicular to the rotational axis is between about 0.5 inches and about 0.7 inches.

In yet another aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The trailing surface defines a portion having a first radius. The bottom surface has a second radius. The first radius is equal to or larger than the second radius. The gullet extends through the first end of the sidewall and defines a central longitudinal axis extending between the

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first and second ends of the sidewall. The central longitudinal axis is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The gullet has a first dimension measured parallel to the rotational axis between the first end of the sidewall and the bottom surface of the gullet that is greater than 1.5 inches. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet. A second dimension measured perpendicular to the rotational axis between the leading surface and a tip of the cutting tooth is between about 0.5 inches and about 0.7 inches.

Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a hole saw according to one embodiment of the invention coupled to a power tool.

FIG. 2 is an end view of the hole saw of FIG. 1.

FIG. 3 is a side view of the hole saw of FIG. 1 in a flattened state and illustrating three gullets.

FIG. 4 is a detailed view of a portion of the hole saw of FIG. 3.

FIG. 5 is a cross sectional view of the hole saw taken along line 5-5 of FIG. 2 illustrating the hole saw cutting through a workpiece.

FIG. 6 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 7 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 8 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 9 is a perspective view of a hole saw according to another embodiment of the invention.

FIG. 10 is a side view of the hole saw of FIG. 9 in a flattened state and illustrating three gullets.

FIG. 11 is a cross sectional view of the hole saw taken along line 11-11 of FIG. 9 with the hole saw cutting through two workpieces.

FIG. 12 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 13 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 14 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 15 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 16 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 17 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 18 is a planar side view of a hole saw according to another embodiment of the invention.

FIG. 19 is a planar side view of a hole saw according to another embodiment of the invention.

DETAILED DESCRIPTION

Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is

to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings. Terms of degree, such as “about,” “substantially,” or “approximately” are understood by those of ordinary skill to refer to reasonable ranges outside of the given value, for example, general tolerances associated with manufacturing, assembly, and use of the described embodiments.

FIGS. 1-5 illustrate a hole saw 100 including a cylindrical body 105 defining a cavity 110. The cylindrical body 105 includes a sidewall 115 having a first end 120 and a second end 125 with the second end 125 positioned opposite the first end 120. The hole saw 100 also includes an end cap 130 coupled to the second end 125 and has an aperture 135 sized to receive an arbor 140 such that the hole saw 100 can be coupled to and driven by a tool 145 (e.g., a power drill or the like) via the arbor 140. The tool 145 drives the hole saw 100 about a rotational axis 150 in a cutting direction 155 (FIG. 2). In the illustrated embodiment, the arbor 140 is selectively coupled to the end cap 130, but in other embodiments, the arbor 140 can be fixedly coupled to the end cap 130. In some embodiments, a pilot drill bit 156 (e.g., a twist drill bit, a spade drill bit, etc.) can be selectively or fixedly coupled to the arbor 140 to extend through the cavity 110 and beyond the first end 120 of the sidewall 115. The illustrated cylindrical body 105 is made of a metallic material (e.g., low carbon steel, etc.). In other embodiments, the cylindrical body 105 can be made of a different metallic material (e.g., high carbon steel, etc.).

As best shown in FIG. 2, the sidewall 115 defines a maximum outer diameter 160. The outer diameter 160 can be between about 1 inch and about 7 inches. In some embodiments, the outer diameter 160 can be about 1.37 inches, about 2.56 inches, about 4 inches, or about 6.25 inches. In further embodiments, the outer diameter 160 can be greater than 7 inches. The sidewall 115 also defines a maximum thickness 165. The thickness 165 can be between about 0.07 inches and about 0.11 inches. In some embodiments, the thickness 165 can be about 0.08 inches, about 0.09 inches, about 0.1 inches, or about 0.104 inches. As best shown in FIG. 3, the sidewall 115 further defines a sidewall height or dimension 170 extending between the first end 120 and the second end 125 in a direction parallel to the rotational axis 150. The illustrated sidewall height 170 is between about 2.5 inches and about 3 inches. In other embodiments, the sidewall height 170 can be between about 1.5 inches and about 5 inches.

In the illustrated embodiment, the sidewall 115 includes three gullets 175 equally spaced about the rotational axis 150 (e.g., each gullet 175 is spaced about 120 degrees relative to an adjacent gullet 175). In other embodiments, the hole saw 100 can include more than three gullets 175 (e.g., four, five, six, etc.) or can include less than three gullets (e.g., two or one) with the gullets 175 equally or non-equally spaced about the rotational axis 150. For example, the hole saw 100 can include two gullets 175 (spaced 180 degrees apart) when the outer diameter 160 is less than about 2.25 inches.

With reference to FIG. 3, each gullet 175 defines a central longitudinal axis 180 extending between the first end 120 and the second end 125. Each gullet 175 includes an open end 185 adjacent the first end 120 and is defined by a bottom surface 190 of the sidewall 115 opposite the open end 185, a leading surface 195 of the sidewall 115 extending between the open end 185 and the bottom surface 190, and a trailing surface 200 of the sidewall 115 extending between the open end 185 and the bottom surface 190. The leading surface 195 and the trailing surface 200 of each gullet 175 are in relation to the cutting direction 155 of the hole saw 100 (i.e., the leading surface 195 is forward the trailing surface 200 as the hole saw 100 moves in the cutting direction 155). Each trailing surface 200 includes a seat or notch 205 formed adjacent the open end 185 and a trailing surface portion 210 positioned between the notch 205 and the bottom surface 190 in a direction parallel to the rotational axis 150. In particular, a first portion 215 of each trailing surface 200 extends between the notch 205 and the trailing surface portion 210 and a second portion 220 of each trailing surface 200 extends between the trailing surface portion 210 and the bottom surface 190. In addition, an edge 225 is positioned between the leading surface 195 and the first end 120 that defines a substantially 90 degree edge. In other embodiments, the edge 225 can be a curved edge.

In the illustrated embodiment, all three gullets 175 include the substantially same geometry, but in other embodiments, two or more gullets 175 can include different geometries. One of the gullets 175 will be described in detail below but can be applicable to one or more of the remaining gullets 175. The illustrated first portion 215 of the trailing surface 200 is a substantially linear surface oriented substantially parallel to the rotational axis 150. In addition, the central longitudinal axis 180 of the gullet 175 is parallel to the rotational axis 150 so that the first portion 215 is also parallel to the central longitudinal axis 180. In other embodiments, the first portion 215 can be a curved surface. The illustrated trailing surface portion 210 defines a first radius 230 between about 1 inch and about 1.25 inches. In other embodiments, the first radius 230 can be between about 1.25 inches and about 2 inches or the first radius 230 can be between about 0.4 inches and about 1 inch. In addition, the trailing surface portion 210 is formed in a lower half of the gullet 175 in a direction parallel to the rotational axis 150. The illustrated second portion 220 of the trailing surface 200 is a substantially linear surface, but in other embodiments, the second portion 220 can be a curved surface. The second portion 220 is also oriented at an oblique angle 235 relative to the rotational axis 150. The illustrated oblique angle 235 is between about 40 degrees and about 50 degrees. In other embodiments, the oblique angle 235 can be between about 50 degrees and about 70 degrees or the oblique angle 235 can be between about 20 degrees and about 40 degrees. A first angle 240 is defined between the first portion 215 and the second portion 220. The illustrated first angle 240 is between about 130 degrees and about 140 degrees. In other embodiments, the first angle 240 can be between 140 degrees and about 170 degrees, or the first angle 240 can be between about 80 degrees and about 130 degrees. In further embodiments, the first radius 230 of the trailing surface portion 210 can be omitted such that only the first angle 240 is formed between the first portion 215 and the second portion 220. The illustrated bottom surface 190 defines a second radius 245 between about 0.25 inches and about 0.4 inches. In other embodiments, the second radius 245 can be between about 0.4 inches and about 1 inch, or the second radius 245 can be between about 0.1 inches and about 0.25

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inches. In some embodiments, the second radius **245** can be equal to or less than the first radius **230**. A ratio of the first radius **230** to the second radius **245** is between about 2.5 to about 5. In other embodiments, the ratio of the first radius **230** to the second radius **245** is between about 1 and about 6. In further embodiments, the ratio of the first radius **230** to the second radius **245** is between about 3 and about 4.

The illustrated leading surface **195** is a substantially linear surface oriented substantially parallel to the rotational axis **150**. A second angle **250** is defined between the second portion **220** and the leading surface **195**. The illustrated second angle **250** is between about 40 degrees and about 50 degrees. In other embodiments, the second angle **250** can be between 50 degrees and about 80 degrees or the second angle **250** can be between about 10 degrees and about 40 degrees. In further embodiments, the second radius **245** can be omitted such that only the second angle **250** is formed between the second portion **220** and the leading surface **195**.

With continued reference to FIG. 3, the gullet **175** defines a gullet height or dimension **255** extending between the first end **120** and the bottom surface **190** in a direction parallel to the rotational axis **150**. The illustrated gullet height **255** is greater than about 1.5 inches. In other embodiments, the gullet height **255** can be between about 1.7 inches and about 1.9 inches, or the gullet height **255** can be between about 1.9 inches and about 3 inches. As such, a ratio of the sidewall height **170** over the gullet height **255** is about 1.4. In other embodiments, the ratio of the sidewall height **170** over the gullet height **255** can be between about 1.1 and about 2. In further embodiments, the ratio of the sidewall height **170** over the gullet height **255** can be between about 2 and about 4. The illustrated sidewall **115** also defines a height **260** extending between the bottom surface **190** of each gullet **175** and the second end **125** of the sidewall **115**. The height **260** is between about 0.7 inches and about 0.9 inches. In other embodiments, the height **260** can be between about 0.9 inches and about 1.5 inches or the height **260** can be between about 0.1 inches and about 0.7 inches. The gullet **175** also defines a maximum gullet width **265** extending between the leading surface **195** and the trailing surface **200** in a direction perpendicular to the rotational axis **150**. In particular, the maximum gullet width **265** extends between the leading surface **195** and a point **270** located on the first portion **215** directly below the notch **205**. The illustrated maximum gullet width **265** is between about 0.75 inches and about 1.2 inches. In other embodiments, the maximum gullet width **265** can be between about 1.2 inches and about 1.5 inches, or the maximum gullet width **265** can be between about 0.4 inches and about 0.75 inches.

The hole saw **100** also includes cutting teeth **275** positioned adjacent the first end **120** of the sidewall **115**. Each cutting tooth **275** has a cutting tip **280** positioned beyond the first end **120** (i.e., above the first end **120** as illustrated in FIG. 4). In particular, the cutting teeth **275** are carbide cutting teeth that are seated within the notches **205** of the sidewall **115** and secured to the sidewall **115** (e.g., by a brazing operation or the like). The carbide cutting teeth **275** include a material hardness that is greater than a material hardness of the sidewall **115**. In other embodiments, the cutting teeth **275** can be made of a different material (e.g., hardened steel, etc.). In further embodiments, the sidewall **115** can form the cutting teeth **275** so that the cutting teeth **275** are not separately secured to the sidewall **115**. Each cutting tooth **275** includes a rake surface **285** and a relief surface **290** with each rake surface **285** facing into one of the gullets **175**. Each rake surface **285** defines a positive rake angle **295** (e.g., angled toward the cutting direction **155**)

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relative to the rotational axis **150** between about 5 degrees and about 15 degrees. As such, a minimum gullet width **296** between each cutting tip **280** and the corresponding leading surface **195** perpendicular to the rotational axis **150** is between about 0.5 inches and about 0.7 inches. In other embodiments, the minimum gullet width **296** can be between about 0.7 inches and about 1.2 inches. In further embodiments, the minimum gullet width **296** can be greater than the maximum gullet width **265**. In yet further embodiments, the rake angle **295** can be a negative rake angle (e.g., angled away from the cutting direction **155**). A tooth angle **300** is defined between the rake surface **285** and the relief surface **290** between about 60 degrees and about 70 degrees (FIG. 4). As best shown in FIG. 2, each cutting tip **280** defines a tip width **305** between about 0.1 inches and about 0.2 inches. The tip width **305** of each cutting tooth **275** is greater than the thickness **165** of the sidewall **115**. As such, a total cutting diameter of the hole saw **100** is greater than the maximum outer diameter **160** of the sidewall **115**. In other embodiments, the tip width **305** can be substantially the same as the thickness **165** of the sidewall **115** with the cutting teeth **275** bent radially inward or outward relative to the rotational axis **150**.

With continued reference to FIG. 4, a recess or relief **310** is formed within the first end **120** of the sidewall **115** directly behind each cutting tooth **275** with a trailing portion **315** of each recess **310** angled in a direction away from the cutting direction **155**. Each recess **310** defines a recess angle **320** relative to the rake surface **285** that is less than the tooth angle **300**. As such, during a grinding process to shape the cutting teeth **275** (e.g., to shape the tooth angle **300**), the recesses **310** provide enough clearance for a grinding wheel or the like to shape the cutting teeth **275** without contacting the sidewall **115**.

Once the cutting teeth **275** are coupled to the sidewall **115**, each gullet **175** defines an area **325** between the rake surface **285**, the trailing surface **200**, the bottom surface **190**, and the leading surface **195**. In the illustrated embodiment, the area **325** of each gullet **175** is equal to or greater than 1.2 inches squared. In other embodiments, the area **325** of each gullet **175** is between about 1.2 inches squared and about 3.5 inches squared. In further embodiments, the area **325** of each gullet **175** is between about 1.2 inches squared and about 2 inches squared. As the hole saw **100** includes three gullets **175**, the total area of the illustrated gullets **175** is equal to or greater than 3.6 inches squared. Each gullet **175** also defines a volume, which is determined by multiplying the area **325** of each gullet **175** by the maximum thickness **165** of the sidewall **115**. As such, the volume of each gullet **175** is equal to or greater than 0.084 inches cubed. In some embodiments, a ratio of the tip width **305** over the volume of one of the gullets **175** is about 0.2. In other embodiments, the ratio of the tip width **305** over the volume of one of the gullets **175** can be between about 0.1 and about 1.1.

With reference to FIG. 5, the hole saw **100** is operable to cut into a workpiece **330** that defines a workpiece height **335**. In the illustrated embodiment, the workpiece height **335** is about 1.5 inches (e.g., height of a standard 2 by 4 piece of lumber). In other embodiments, the workpiece height **335** can be less than 1.5 inches. In operation, the cutting teeth **275** cut into the workpiece **330**, thereby forming workpiece chips that collect within the gullets **175**. As the hole saw **100** bores deeper into the workpiece **330**, more workpiece chips are collected within the gullets **175**. Accordingly, the illustrated gullets **175** include a particular volume to accommodate the workpiece chips as the hole saw **100** cuts into the workpiece **330**. If the volume of the gullets

175 is not sufficient to accommodate the workpiece chips, the workpiece chips will build up within the gullets 175 and decrease the efficiency of the hole saw 100 cutting into the workpiece 330. Once the hole saw 100 bores through the workpiece 330, a cylindrical workpiece plug will be removed from the cavity 110 of the hole saw 100. In particular, a user can grip the cylindrical workpiece plug (via their fingers) through the gullets 175 to axially slide the cylindrical workpiece plug along the rotational axis and out of the cavity 110. The minimum gullet width 296 of each gullet 175 is sized to provide enough clearance between each cutting tip 280 and the corresponding leading surface 195 so that a user's finger avoids contact with the cutting tip 280 as the cylindrical workpiece plug is removed from the hole saw 100.

FIG. 6 illustrates a planar view of a portion of a hole saw 400 according to another embodiment. The hole saw 400 is similar to the hole saw 100 of FIGS. 1-5; therefore, similar components are designated with similar reference numbers plus 300. At least some differences and/or at least some similarities between the hole saws 100, 400 will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

The illustrated hole saw 400 is moveable about a rotational axis 450 in a cutting direction 455 and includes a sidewall 415 having a first end 420 and a second end 425. The sidewall 415 also includes gullets 475 (only one gullet 475 is illustrated in FIG. 6) each defining a central longitudinal axis 480. Each gullet 475 includes a notch 505, an open end 485, a bottom surface 490 defining a second radius 545, a leading surface 495, and a trailing surface 500. Each trailing surface 500 includes a trailing surface portion 510 defining a first radius 530, a first portion 515, and a second portion 520 with the trailing surface portion 510 positioned between the first portion 515 and the second portion 520. In the illustrated embodiment, each first portion 515 is parallel to the leading surface 495; however, in other embodiments, the first portion 515 can be obliquely angled relative to the leading surface 495. In other embodiments, the second portion 520 can be omitted so that the trailing surface portion 510 directly connects with the bottom surface 490. In the illustrated embodiment, the first portion 515 of each trailing surface 500 is oriented at a third angle 640 between about 5 degrees and about 15 degrees relative to the rotational axis 450 so that the first portion 515 is angled from the first end 420 to the second end 425 in a direction toward the cutting direction 455. As a result, each central longitudinal axis 480 is also oriented between about 5 degrees and about 15 degrees relative to the rotational axis 450 so that the central longitudinal axis 480 is angled from the first end 420 to the second end 425 in a direction toward the cutting direction 455. In other embodiments, the third angle 640 is greater than about 5 degrees. In further embodiments, the third angle 640 is less than about 15 degrees. In yet further embodiments, the third angle 640 is between about 1 degree and about 40 degrees.

The hole saw 400 also includes cutting teeth 575 (only one cutting tooth 575 is illustrated in FIG. 6) each having a cutting tip 580 and seated within one notch 505. Each cutting tooth 575 includes a rake surface 585 and a relief surface 590.

FIG. 7 illustrates a planar view of a portion of a hole saw 700 according to another embodiment. The hole saw 700 is similar to the hole saw 100 of FIGS. 1-5; therefore, similar

components are designated with similar reference numbers plus 600. At least some differences and/or at least some similarities between the hole saws 100, 700 will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

The illustrated hole saw 700 is moveable about a rotational axis 750 in a cutting direction 755 and includes a sidewall 715 having a first end 720 and a second end 725. The sidewall 715 also includes gullets 775 (only one gullet 775 is illustrated in FIG. 7) each defining a central longitudinal axis 780. Each gullet 775 includes a notch 805, an open end 785, a bottom surface 790 defining a second radius 845, a leading surface 795, and a trailing surface 800. Each trailing surface 800 includes a trailing surface portion 810 defining a first radius 830, a first portion 815, and a second portion 820 with the trailing surface portion 810 positioned between the first portion 815 and the second portion 820. Each leading surface 795 includes a third portion 945, a fourth portion 950, and a support surface 955 positioned between the third portion 945 and the fourth portion 950 in a direction parallel to the rotational axis 750. The support surface 955 is also positioned at a support surface distance 960 between about 0.7 inches and about 1 inch from the first end 720 of the sidewall 715. In other embodiments, the support surface 955 can be formed within at least one of the gullets 775. In the illustrated embodiment, the third portion 945 is oriented parallel to the fourth portion 950 of each leading surface 795 and the first portion 815 of each trailing surface 800. Each third portion 945 and fourth portion 950 are also offset (e.g., nonlinear) relative to each other. In other embodiments, the third portion 945 can be obliquely angled relative to the first portion 815 and/or the fourth portion 950. Each support surface 955 is oriented at a support angle 965 relative to the rotational axis 750. The support angle 965 is less than 90 degrees relative to the rotational axis 750 so that the support surface 955 is angled downwardly and toward the cutting direction 755 (e.g., the support surface 955 generally faces upwardly toward the first end 720). In other embodiments, the support angle 965 can be less than 90 degrees and greater than 75 degrees relative to the rotational axis 750. Also, each support surface 955 is positioned above the trailing surface portion 810 in a direction parallel to the rotational axis 750. In other embodiments, each support surface 955 can be positioned at the same height of the trailing surface portion 810 or positioned below the trailing surface portion 810.

The hole saw 700 also includes cutting teeth 875 (only one cutting tooth 875 is illustrated in FIG. 7) each having a cutting tip 880 and seated within one notch 805. Each cutting tooth 875 includes a rake surface 885 and a relief surface 890.

In operation, a cylindrical workpiece plug can be removed from the hole saw 700 by using a tool (e.g., a screwdriver or the like). In particular, the tool is inserted within one of the gullets 775 below the cylindrical workpiece plug. In some embodiments, the tool engages the bottom surface 790 of the gullet 775 to leverage and push the cylindrical workpiece plug out of the first end 720. The tool can also engage the support surface 955 to leverage and push the cylindrical workpiece plug out of the first end 720. As the support surface 955 is angled downwardly and toward the cutting direction 755, the tool can be captured on the support surface

955 without sliding into the gullet **775** as the tool is used to leverage the cylindrical workpiece plug out of the hole saw **700**.

FIG. **8** illustrates a planar view of a portion of a hole saw **1000** according to another embodiment. The hole saw **1000** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus **900**. At least some differences and/or at least some similarities between the hole saws **100**, **1000** will be discussed in detail below. In addition, components or features described with respect to only one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

The illustrated hole saw **1000** is moveable about a rotational axis **1050** in a cutting direction **1055** and includes a sidewall **1015** having a first end **1020** and a second end **1025**. The sidewall **1015** also includes gullets **1075** (only one gullet **1075** is illustrated in FIG. **8**) each defining a central longitudinal axis **1080**. Each gullet **1075** includes a notch **1105**, an open end **1085**, a bottom surface **1090** defining a second radius **1145**, a leading surface **1095**, and a trailing surface **1100**. Each trailing surface **1100** includes a trailing surface portion **1110** defining a first radius **1130**, a first portion **1115**, and a second portion **1120** with the trailing surface portion **1110** positioned between the first portion **1115** and the second portion **1120**. Each leading surface **1095** includes a third portion **1245**, a fourth portion **1250**, and a support surface **1255** (similar to the support surface **955** of FIG. **7**) formed on a protrusion **1270** of the sidewall **1015**, which extends into each gullet **1075**. The illustrated third portion **1245** and fourth portion **1250** are substantially collinear surfaces with the support surface **1255** oriented at a substantially 90 degree angle relative to the rotational axis **1050**.

The hole saw **1000** also includes cutting teeth **1175** (only one cutting tooth **1175** is illustrated in FIG. **8**) each having a cutting tip **1180** and seated within one notch **1105**. Each cutting tooth **1175** includes a rake surface **1185** and a relief surface **1190**.

FIGS. **9-11** illustrate a hole saw **1300** according to another embodiment. The hole saw **1300** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus **1200**. At least some differences and/or at least some similarities between the hole saws **100**, **1300** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

The illustrated hole saw **1300** is moveable about a rotational axis **1350** in a cutting direction **1355** and includes a cylindrical body **1305** having a sidewall **1315** that defines a cavity **1310** and an end cap **1330** having an aperture **1335**. The sidewall **1315** includes a first end **1320** and a second end **1325**. The sidewall **1315** defines a maximum outer diameter **1360** (FIG. **11**), a maximum thickness **1365** (FIG. **11**), and a sidewall height or dimension **1370** (FIG. **10**) extending between the first end **1320** and the second end **1325** in a direction parallel to the rotational axis **1350**. The sidewall height **1370** can be between about 3.3 inches and about 3.7 inches. In other embodiments, the sidewall height **1370** can be between about 3.7 inches and about 4.5 inches. The sidewall **1315** also includes gullets **1375** each defining a central longitudinal axis **1380** oriented at an oblique angle (e.g., between about 5 degrees and about 15 degrees) relative to the rotational axis **1350**. Each gullet **1375** includes a notch

1405, an open end **1385**, a bottom surface **1390** defining a second radius **1445**, a leading surface **1395**, and a trailing surface **1400**. Each trailing surface **1400** includes a trailing surface portion **1410** defining a first radius **1430**, a first portion **1415**, and a second portion **1420** with the trailing surface portion **1410** positioned between the first portion **1415** and the second portion **1420**. Each leading surface **1395** includes an edge **1425** and two support surfaces **1555a**, **1555b** (similar to the support surfaces **955**, **1255** illustrated in FIGS. **7** and **8**) so that each leading surface **1395** includes a third portion **1545**, a fourth portion **1550**, and a fifth portion **1575**. With the two support surfaces **1555a**, **1555b** formed in the leading surface **1395**, the leading surface **1395** defines a zig-zag shaped surface. The illustrated third portion **1545**, fourth portion **1550**, and fifth portion **1575** are all linear surfaces with the third portion **1545** obliquely oriented relative to the fourth portion **1550** and the fourth portion **1550** oriented substantially parallel to the fifth portion **1575**. In other embodiments, two or more of the third portion **1545**, the fourth portion **1550**, and the fifth portion **1575** can be substantially parallel relative to each other or obliquely oriented relative to each other. In further embodiments, at least one of the third portion **1545**, the fourth portion **1550**, and the fifth portion **1575** can be a curved surface. In yet further embodiments, the third portion **1545**, the fourth portion **1550**, the fifth portion **1575**, and the two support surface **1555a**, **1555b** can be omitted so that the leading surface **1395** is one linear surface. The illustrated first support surface **1555a** is positioned from the first end **1320** at a first support distance **1560a** of between about 2 inches and about 2.2 inches. The second support surface **1555b** is positioned from the first end **1320** at a second support distance **1560b** of between about 1 inch and about 1.5 inches. In other embodiments, each leading surface **1395** can include one support surface. In other embodiments, each leading surface **1395** can include more than two support surfaces.

In addition, each gullet **1375** defines an oblique angle **1435** between the second portion **1420** and the rotational axis **1350**. The oblique angle **1435** is between about 55 degrees and about 65 degrees. In particular, the oblique angle **1435** is defined between the rotational axis **1350** and a tangent line extending through a midpoint of the second portion **1420**. A first angle **1440** is defined between the first portion **1415** and the second portion **1420** that is between about 120 degrees and about 135 degrees. In particular, the first angle **1440** is defined between the first portion **1415** and a tangent line extending through a midpoint of the second portion **1420**. In other embodiments with the first portion **1415** defining a curved surface, the first angle **1440** is defined between tangent lines extending through midpoints of the first and second portions **1415**, **1420**. A second angle **1450** is defined between the second portion **1420** and the third portion **1545** that is between about 35 degrees and about 45 degrees. In particular, the second angle **1450** is defined between the third portion **1545** and a tangent line extending through a midpoint of the second portion **1420**. In other embodiments with the third portion **1545** defining a curved surface, the second angle **1450** is defined between tangent lines extending through midpoints of the second and third portions **1420**, **1545**.

As best shown in FIG. **10**, each gullet **1375** defines a gullet height or dimension **1455** extending between the first end **1320** and the bottom surface **1390** in a direction parallel to the rotational axis **1350**. The illustrated gullet height **1455** is between about 3 inches and about 3.2 inches. In other embodiments, the gullet height **1455** can be between about

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3.2 inches and about 4 inches, or the gullet height **1455** can be between about 2.5 inches and about 3 inches. As such, a ratio of the sidewall height **1370** over the gullet height **1455** is between about 1.05 and about 1.2. In other embodiments, the ratio can be between about 1.2 and about 2. The illustrated sidewall **1315** also defines a height **1460** extending between the bottom surface **1390** of each gullet **1375** and the second end **1325** of the sidewall **1315**. The height **1460** is between about 0.3 inches and about 0.4 inches. In other embodiments, the height **1460** can be between about 0.4 inches and about 1 inch, or the height **1460** can be between about 0.1 inches and about 0.3 inches. In addition, a maximum gullet width **1465** is defined by a point **1470** on the first portion **1415** and the fifth portion **1575** adjacent the second support surface **1555b**.

The hole saw **1300** also includes cutting teeth **1475** each having a cutting tip **1480** and seated within one notch **1405**. Each cutting tooth **1475** includes a rake surface **1485** and a relief surface **1490**. A minimum gullet width **1496** extends between each cutting tip **1480** and the corresponding leading surface **1395**. Once the cutting teeth **1475** are coupled to the sidewall **1315**, each gullet **1375** defines an area **1525** between the rake surface **1485**, the trailing surface **1400**, the bottom surface **1390**, and the leading surface **1395**. In the illustrated embodiment, the area **1525** of each gullet **1375** is equal to or greater than 3.2 inches squared. In other embodiments, the area **1525** of each gullet **1375** is between about 2 inches squared and about 4 inches squared. In further embodiments, the area **1525** of each gullet **1375** is between about 3.2 inches squared and about 4 inches squared. As the hole saw **1300** includes three gullets **1375**, the total area of the illustrated gullets **1375** is equal to or greater than 9.6 inches squared. Each gullet **1375** also defines a volume, which is determined by multiplying the area **1525** of each gullet **1375** by the maximum thickness **1365** of the sidewall **1315**. As such, the volume of each gullet **1375** is equal to or greater than 0.256 inches cubed. In other embodiments, the volume of each gullet **1375** can be between about 0.227 inches cubed and about 0.3 inches cubed.

With reference to FIG. 11, the hole saw **1300** is operable to cut into two workpieces **1530** that defines a workpiece height **1535**. In the illustrated embodiment, the workpiece height **1535** is about 3 inches (e.g., two standard 2 by 4 pieces of lumber). In other embodiments, the workpiece height **1535** can be less than 3 inches. In further embodiments, the workpiece height **1535** can be between about 1.5 inches and about 3 inches. In yet further embodiments, the two workpieces **1530** can be formed as one workpiece with the workpiece height **1535**. In operation, the cutting teeth **1475** cut into the two workpieces **1530**, thereby forming workpiece chips that collect within the gullets **1375**. As the hole saw **1300** bores deeper into the two workpieces **1530**, more workpiece chips are collected within the gullets **1375**. Accordingly, the illustrated gullets **1375** include a particular volume to accommodate the workpiece chips as the hole saw **1300** cuts into the two workpieces **1530**. Once the hole saw **1300** bores through the two workpieces **1530**, a cylindrical workpiece plug will be removed from the cavity **1310** of the hole saw **1300**. In one embodiment, a user can grip the cylindrical workpiece plug (via their fingers) through the gullets **1375** to axially slide the cylindrical workpiece plug along the rotational axis **1350** and out of the cavity **1310**. In other embodiments, a tool can be inserted into one of the gullets **1375** to be supported on one of the support surfaces **1555a**, **1555b** for a user to leverage the cylindrical workpiece plug out of the hole saw **1300**.

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FIG. 12 illustrates a planar view of a portion of a hole saw **1600** according to another embodiment. The hole saw **1600** is similar to the hole saw **100** of FIGS. 1-5; therefore, similar components are designated with similar references numbers plus 1500. At least some differences and/or at least some similarities between the hole saws **100**, **1600** will be discussed in detail below. In addition, components or features described with respect to only one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **1600** can be sized to bore into the single workpiece **330** of FIG. 5 similar to the hole saw **100**, or the hole saw **1600** can be sized to bore into the two workpieces **1530** of FIG. 11 similar to the hole saw **1300**.

The illustrated hole saw **1600** is moveable about a rotational axis **1650** in a cutting direction **1655** and includes a sidewall **1615** having a first end **1620** and a second end **1625**. The sidewall **1615** also includes gullets **1675** (only one gullet **1675** is illustrated in FIG. 12) each defining a central longitudinal axis **1680**. Each gullet **1675** includes a notch **1705**, an open end **1685**, a bottom surface **1690** defining a second radius **1745**, a leading surface **1695**, and a trailing surface **1700**. Each trailing surface **1700** includes a trailing surface portion **1710** defining a first radius **1730**, a first portion **1715**, and a second portion **1720** with the trailing surface portion **1710** positioned between the first portion **1715** and the second portion **1720**. The sidewall **1615** also includes three support apertures **1880** (only one is illustrated in FIG. 12) each positioned at a substantially same location that is circumferentially between adjacent gullets **1675**. In particular, each support aperture **1880** is positioned above the bottom surfaces **1690** of the gullets **1675**. For example, each support aperture **1880** is positioned above the trailing surface portions **1710** of the gullets **1675**. Each support aperture **1880** is also positioned between adjacent gullets **1675** as to not circumferentially overlap with any portion of the gullets **1675**. Each support aperture **1880** is oblong-shaped and includes a longitudinal axis that is substantially perpendicular to the rotational axis **1650**. The support apertures **1880** function similar to the support surfaces illustrated within FIGS. 7-11 for a tool to be inserted into one of the support apertures **1880** to leverage and remove a cylindrical workpiece plug from the hole saw **1600**. In other embodiments, the sidewall **1615** can include fewer than or more than three support apertures **1880** and/or the support apertures **1880** can be located at different positions circumferentially around the sidewall **1615**.

The hole saw **1600** also includes cutting teeth **1775** (only one cutting tooth **1775** is illustrated in FIG. 12) each having a cutting tip **1780** and seated within one notch **1705**. Each cutting tooth **1775** includes a rake surface **1785** and a relief surface **1790**.

FIG. 13 illustrates a planar view of a portion of a hole saw **1900** according to another embodiment. The hole saw **1900** is similar to the hole saw **100** of FIGS. 1-5; therefore, similar components are designated with similar references numbers plus 1800. At least some differences and/or at least some similarities between the hole saws **100**, **1900** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **1900** can be sized to bore into the single workpiece **330** of FIG. 5 similar to the

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hole saw 100, or the hole saw 1900 can be sized to bore into the two workpieces 1530 of FIG. 11 similar to the hole saw 1300.

The illustrated hole saw 1900 is moveable about a rotational axis 1950 in a cutting direction 1955 and includes a sidewall 1915 having a first end 1920 and a second end 1925. The sidewall 1915 also includes gullets 1975 (only one gullet 1975 is illustrated in FIG. 13) each defining a central longitudinal axis 1980. Each gullet 1975 includes a notch 2005, an open end 1985, a bottom surface 1990 defining a second radius 2045, a leading surface 1995, and a trailing surface 2000. In particular, the second radius 2045 of each bottom surface 1990 extends directly from the trailing surface 2000 to the leading surface 1995. The sidewall 1915 also includes three support apertures 2180 (only one is illustrated in FIG. 13) each positioned below the bottom surfaces 1990. Each support aperture 2180 is also positioned as to circumferentially overlap with one of the gullets 1975.

The hole saw 1900 also includes cutting teeth 2075 (only one cutting tooth 2075 is illustrated in FIG. 13) each having a cutting tip 2080 and seated within one notch 2005. Each cutting tooth 2075 includes a rake surface 2085 and a relief surface 2090.

FIG. 14 illustrates a planar view of a portion of a hole saw 2200 according to another embodiment. The hole saw 2200 is similar to the hole saw 100 of FIGS. 1-5; therefore, similar components are designated with similar reference numbers plus 2100. At least some differences and/or at least some similarities between the hole saws 100, 2200 will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw 2200 can be sized to bore into the single workpiece 330 of FIG. 5 similar to the hole saw 100, or the hole saw 2200 can be sized to bore into the two workpieces 1530 of FIG. 11 similar to the hole saw 1300.

The illustrated hole saw 2200 is moveable about a rotational axis 2250 in a cutting direction 2255 and includes a sidewall 2215 having a first end 2220 and a second end 2225. The sidewall 2215 also includes gullets 2275 (only one gullet 2275 is illustrated in FIG. 14) each defining a central longitudinal axis 2280. Each gullet 2275 includes a notch 2305, an open end 2285, a bottom surface 2290 defining a second radius 2345, a leading surface 2295, and a trailing surface 2300. Each trailing surface 2300 includes a trailing surface portion 2310 defining a first radius 2330, a first portion 2315, and a second portion 2320 with the trailing surface portion 2310 positioned between the first portion 2315 and the second portion 2320. The first portion 2315 of each gullet 2275 defines a third angle 2440 that extends from the notch 2305 to the trailing surface portion 2310 in a direction away from the cutting direction 2255. As a result, the trailing surface portion 2310 is positioned behind the first portion 2315 relative to the cutting direction 2255. In other embodiments, the third angle 2440 can be oriented in a direction toward the cutting direction 2255 with at least a portion of the trailing surface portion 2310 positioned behind a portion of the first portion 2315 in a direction relative to the cutting direction 2255. The sidewall 2215 also includes three support apertures 2480 (only one is illustrated in FIG. 14) each positioned below a trailing surface portion 2310.

The hole saw 2200 also includes cutting teeth 2375 (only one cutting tooth 2375 is illustrated in FIG. 14) each having

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a cutting tip 2380 and seated within one notch 2305. Each cutting tooth 2375 includes a rake surface 2385 and a relief surface 2390.

FIG. 15 illustrates a planar view of a portion of a hole saw 2500 according to another embodiment. The hole saw 2500 is similar to the hole saw 100 of FIGS. 1-5; therefore, similar components are designated with similar reference numbers plus 2400. At least some differences and/or at least some similarities between the hole saws 100, 2500 will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw 2500 can be sized to bore into the single workpiece 330 of FIG. 5 similar to the hole saw 100, or the hole saw 2500 can be sized to bore into the two workpieces 1530 of FIG. 11 similar to the hole saw 1300.

The illustrated hole saw 2500 is moveable about a rotational axis 2550 in a cutting direction 2555 and includes a sidewall 2515 having a first end 2520 and a second end 2525. The sidewall 2515 also includes gullets 2575 (only one gullet 2575 is illustrated in FIG. 15) each defining a central longitudinal axis 2580. Each gullet 2575 includes a notch 2605, an open end 2585, a bottom surface 2590 defining a second radius 2645, a leading surface 2595, and a trailing surface 2600. Each leading surface 2595 includes a support surface 2755 positioned between a third portion 2745 and a fourth portion 2750.

The hole saw 2500 also includes cutting teeth 2675 (only one cutting tooth 2675 is illustrated in FIG. 15) each having a cutting tip 2680 and seated within one notch 2605. Each cutting tooth 2675 includes a rake surface 2685 and a relief surface 2690.

FIG. 16 illustrates a planar view of a portion of a hole saw 2800 according to another embodiment. The hole saw 2800 is similar to the hole saw 100 of FIGS. 1-5; therefore, similar components are designated with similar reference numbers plus 2700. At least some differences and/or at least some similarities between the hole saws 100, 2800 will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw 2800 can be sized to bore into the single workpiece 330 of FIG. 5 similar to the hole saw 100, or the hole saw 2800 can be sized to bore into the two workpieces 1530 of FIG. 11 similar to the hole saw 1300.

The illustrated hole saw 2800 is moveable about a rotational axis 2850 in a cutting direction 2855 and includes a sidewall 2815 having a first end 2820 and a second end 2825. The sidewall 2815 also includes gullets 2875 (only one gullet 2875 is illustrated in FIG. 16) each defining a central longitudinal axis 2880. Each gullet 2875 includes a notch 2905, an open end 2885, a bottom surface 2890 defining a second radius 2945, a leading surface 2895, and a trailing surface 2900. Each trailing surface 2900 includes a trailing surface portion 2910 defining a first radius 2930, a first portion 2915, and a second portion 2920 with the trailing surface portion 2910 positioned between the first portion 2915 and the second portion 2920. The sidewall 2815 also includes three support apertures 3080 (only one is illustrated in FIG. 16). Each support aperture 3080 defines a greater area than an area of one of the gullets 2875. Each support aperture 3080 circumferentially overlaps with one of the gullets 2875 so that the leading surface 2895 of the gullet

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2875 is positioned forward the support aperture **3080** but a portion of the support aperture **3080** is positioned behind the trailing surface **2900** of the gullet **2875** relative to the cutting direction **2855**.

The hole saw **2800** also includes cutting teeth **2975** (only one cutting tooth **2975** is illustrated in FIG. **16**) each having a cutting tip **2980** and seated within one notch **2905**. Each cutting tooth **2975** includes a rake surface **2985** and a relief surface **2990**.

FIG. **17** illustrates a planar view of a portion of a hole saw **3100** according to another embodiment. The hole saw **3100** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus **3000**. At least some differences and/or at least some similarities between the hole saws **100**, **3100** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3100** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100** or the hole saw **3100** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

The illustrated hole saw **3100** is moveable about a rotational axis **3150** in a cutting direction **3155** and includes a sidewall **3115** having a first end **3120** and a second end **3125**. The sidewall **3115** also includes gullets **3175** (only one gullet **3175** is illustrated in FIG. **17**) each defining a central longitudinal axis **3180**. Each gullet **3175** includes a notch **3205**, an open end **3185**, a bottom surface **3190** defining a second radius **3245**, a leading surface **3195**, and a trailing surface **3200**. Each trailing surface **3200** includes a trailing surface portion **3210** defining a first radius **3230**, a first portion **3215**, and a second portion **3220** with the trailing surface portion **3210** positioned between the first portion **3215** and the second portion **3220**. The sidewall **3115** also includes three support apertures **3380** (only one is illustrated in FIG. **17**). Each support aperture **3380** circumferentially overlaps with one of the gullets **3175** so that a portion of the support aperture **3380** is positioned forward the leading surface **3195** but a portion of the first portion **3215** is positioned behind the support aperture **3380** relative to the cutting direction **3155**.

The hole saw **3100** also includes cutting teeth **3275** (only one cutting tooth **3275** is illustrated in FIG. **17**) each having a cutting tip **3280** and seated within one notch **3205**. Each cutting tooth **3275** includes a rake surface **3285** and a relief surface **3290**.

FIG. **18** illustrates a planar view of a portion of a hole saw **3400** according to another embodiment. The hole saw **3400** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus **3300**. At least some differences and/or at least some similarities between the hole saws **100**, **3400** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3400** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **3400** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

The illustrated hole saw **3400** is moveable about a rotational axis **3450** in a cutting direction **3455** and includes a sidewall **3415** having a first end **3420** and a second end

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3425. The sidewall **3415** also includes three pairs of gullets **3475** (only one pair of gullets **3475** is illustrated in FIG. **18**). Each pair **3475** includes a smaller gullet **3475a** defining a central longitudinal axis **3480a** and a larger gullet **3475b** defining a central longitudinal axis **3480b**. In the illustrated embodiment, the central longitudinal axes **3480a**, **3480b** are substantially parallel; however, in other embodiments, the central longitudinal axes **3480a**, **3480b** can be obliquely angled. The smaller gullet **3475a** includes a notch **3505a**, an open end **3485a**, a bottom surface **3490a** defining a second radius **3545a**, a leading surface **3495a**, and a trailing surface **3500a**. Likewise, the larger gullet **3475b** includes a notch **3505b**, an open end **3485b**, a bottom surface **3490b** defining a second radius **3545b**, a leading surface **3495b**, and a trailing surface **3500b**. The bottom surface **3490b** of the larger gullet **3475b** is positioned closer to the second end **3425** of the sidewall **3415** than the bottom surface **3490a** of the smaller gullet **3475a**. In other embodiments, the sidewall **3415** can include fewer or more than three pairs of gullets **3475**. In the illustrated embodiment, the second radius **3545b** of the larger gullet **3475b** is greater than the second radius **3545a** of the smaller gullet **3475a**. In other embodiments, the second radius **3545b** of the larger gullet **3475b** can be equal to or less than the second radius **3545a** of the smaller gullet **3475a**.

The hole saw **3400** also includes cutting teeth **3575** (only one pair of cutting teeth **3575** are illustrated in FIG. **18**) each having a cutting tip **3580** and seated within one notch **3505a**, **3505b**. Each cutting tooth **3575** includes a rake surface **3585** and a relief surface **3590**.

FIG. **19** illustrates a planar view of a portion of a hole saw **3700** according to another embodiment. The hole saw **3700** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus **3600**. At least some differences and/or at least some similarities between the hole saws **100**, **3700** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3700** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **3700** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

The illustrated hole saw **3700** is moveable about a rotational axis **3750** in a cutting direction **3755** and includes a sidewall **3715** having a first end **3720** and a second end **3725**. The sidewall **3715** also includes gullets **3775** (only one gullet **3775** is illustrated in FIG. **19**) each defining a central longitudinal axis **3780** parallel to the rotational axis **3750**. Each gullet **3775** includes a notch **3805**, an open end **3785**, a bottom surface **3790** defining a second radius **3845**, a leading surface **3795**, and a trailing surface **3800**. Each trailing surface **3800** includes a trailing surface portion **3810** defining a first radius **3830**, a first portion **3815**, and a second portion **3820** with the trailing surface portion **3810** positioned between the first portion **3815** and the second portion **3820**. The sidewall **3715** also includes three support apertures **3980** (only one is illustrated in FIG. **19**). Each support aperture **3980** defines a longitudinal axis that substantially aligns with one central longitudinal axis **3780** of one gullet **3775**. In particular, each support aperture **3980** is directly below one gullet **3775**.

The hole saw **3700** also includes cutting teeth **3875** (only one cutting tooth **3875** is illustrated in FIG. **19**) each having

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a cutting tip **3880** and seated within one notch **3805**. Each cutting tooth **3875** includes a rake surface **3885** and a relief surface **3890**.

Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described. Various features and advantages of the invention are set forth in the following claims.

What is claimed is:

1. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising:

a cylindrical body including

a sidewall having a first end and a second end opposite the first end,

a gullet formed in the sidewall between a leading surface, a trailing surface, and a linear bottom surface of the sidewall, the gullet extending through the first end of the body, and

a support aperture formed in the sidewall, the support aperture entirely surrounded by the sidewall, at least one side of the support aperture being parallel to the linear bottom surface of the sidewall, and an entirety of the support aperture being positioned closer than the linear bottom surface of the sidewall to the second end of the sidewall;

a cutting tooth coupled to the first end of the sidewall adjacent the gullet; and

an end cap coupled to the second end of the sidewall.

2. The hole saw of claim 1, wherein the support aperture has a maximum width measured in a direction perpendicular to the rotational axis, and wherein the maximum width of the support aperture is substantially equal to a maximum width of the gullet.

3. The hole saw of claim 1, wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than half of a height of the sidewall.

4. The hole saw of claim 1, wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than a height of the support aperture.

5. The hole saw of claim 1, wherein the support aperture is at least partially defined by four linear surfaces of the sidewall.

6. The hole saw of claim 1, wherein a shortest distance between the gullet and the support aperture measured parallel to the rotational axis is less than a shortest distance between the support aperture and the second end of the sidewall measured parallel to the rotational axis.

7. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising:

a cylindrical body including

a sidewall having a first end and a second end opposite the first end,

a gullet formed in the sidewall between a first surface, a second surface, and a third surface of the sidewall, the third surface extending between the first and second surfaces, the gullet extending through the first end of the sidewall, and

a polygonal opening formed in the sidewall, the polygonal opening entirely surrounded by the sidewall and positioned between the third surface and the second end of the sidewall;

a cutting tooth coupled to the first end of the sidewall adjacent the second surface of the gullet; and

an end cap coupled to the second end of the sidewall,

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wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than a height of the polygonal opening.

8. The hole saw of claim 7, wherein the polygonal opening has a rounded parallelogram shape.

9. The hole saw of claim 7, wherein the polygonal opening includes a first side, a second side, a third side, and a fourth side, and wherein the first side of the polygonal opening is parallel with the third surface of the sidewall.

10. The hole saw of claim 9, wherein the first and second sides of the polygonal opening are parallel, and wherein the third and fourth sides of the polygonal opening are parallel.

11. The hole saw of claim 9, wherein the first, second, third, and fourth sides of the polygonal opening are connected by rounded corners.

12. The hole saw of claim 9, wherein the polygonal opening has a height measured between the first and second sides and parallel to the rotational axis, and wherein the height of the polygonal opening is greater than a shortest distance between the third surface of the sidewall and the first side of the polygonal opening measured parallel to the rotational axis.

13. The hole saw of claim 9, wherein the third surface of the sidewall is linear, and wherein the first side of the polygonal opening is linear.

14. The hole saw of claim 13, wherein a first arcuate surface is positioned between the first and third surfaces of the sidewall, and wherein a second arcuate surface is positioned between the second and third surfaces of the sidewall.

15. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising:

a cylindrical body including

a sidewall having a first end and a second end opposite the first end,

a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall, the gullet extending through the first end of the sidewall, the gullet having a maximum width measured between the leading surface and the trailing surface in a direction perpendicular to the rotational axis, and

a support aperture formed in the sidewall, the support aperture having a first side, a second side opposite the first side, a linear third side extending between the first side and the second side, and a maximum width measured between the first side and the second side in the direction perpendicular to the rotational axis, the maximum width of the support aperture being substantially equal to the maximum width of the gullet;

a cutting tooth coupled to the first end of the sidewall adjacent the gullet; and

an end cap coupled to the second end of the sidewall.

16. The hole saw of claim 15, wherein the support aperture also has a fourth side extending between the first side and the second side.

17. The hole saw of claim 16, wherein the third side of the support aperture is a side of the support aperture nearest the gullet, the fourth side of the support aperture is a side of the support aperture nearest the second end of the sidewall, and the support aperture is devoid of material between the third and fourth sides.

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18. The hole saw of claim **15**, wherein the third side of the support aperture is positioned closer than the bottom surface of the sidewall to the second end of the sidewall.

19. The hole saw of claim **15**, wherein
the gullet has a height measured parallel to the rotational axis,
the support aperture has a height measured parallel to the rotational axis, and
the height of the gullet is greater than the height of the support aperture.

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20. The hole saw of claim **15**, wherein a width of the gullet narrows from the maximum width as the gullet approaches the first end of the sidewall.

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